A formal approach to chart patterns classification in financial time series

Yuqing Wan[a], Yain-Whar Si[a,1,*]

[a]*Department of Computer and Information Science, University of Macau, Macau, China*

Abstract

Classifying chart patterns from input subsequences is a crucial pre-processing step in technical analysis. In this paper, we compile comprehensive formal specifications of 53 chart patterns reported in the literature. A first-order logic representation is chosen to describe the shape and corresponding constraints of each pattern. These formal specifications are formulated in such a way that data mining algorithms can use them for classification without significant modification. These formal specifications are also intended to serve as a reference model for future research in the chart patterns classification area. Using these formal specifications, we perform extensive experiments using a real dataset from the NYSE Composite (NYSE). The performance of the proposed method is compared against Template Based (TB), Euclidean Distance (ED), and Dynamic Time Warping (DTW) approaches. The experimental results show that the rules translated from the specifications can be effectively used to identify chart patterns from real datasets.

Keywords: chart patterns, formal specification, pattern matching, financial time series, technical analysis

1. Introduction

In technical analysis, the appearance of chart patterns in financial time series is often considered to be a signal of triggering a trade. The detected patterns are also used to predict future price trends. Various pattern matching approaches can be used to measure the similarities between chart pattern templates and the subsequences extracted from financial time series. For instance, the template-based (TB) [11] approach measures the temporal and amplitude distance between a chart pattern template and the query subsequence. The rule-based (RB) [11] approach relies on a set of predefined rules for classifying the query subsequence. Euclidean distance (ED) and dynamic time warping (DTW) [4] measure the distance between a re-scaled chart pattern template and the original subsequence. Regardless of the choice of methods or similarity measures used for pattern matching, there is currently no industry-wide standard for how these patterns should be defined in an unambiguous way. Some of the well-known patterns, such as "Head-and-Shoulders", "Triple Tops" and "Cup with Handle" are frequently used to evaluate the effectiveness of various pattern

*Corresponding author

Email addresses: qinggor@qq.com (Yuqing Wan), fstasp@umac.mo (Yain-Whar Si)

[1]Faculty of Science and Technology, University of Macau, E11-4022, Avenida da Universidade, Taipa, Macau, China, Tel.: +853 8822-4454

matching methods. However, there is no documented work on the precise definitions of these chart patterns. Although the shapes of 53 known chart patterns and their usefulness in predicting future price trends have been documented in ([5],[9]), these reports are written in natural language. Therefore, it is certainly possible that the shape of one pattern may differ slightly from one definition to another. As there is no agreed-upon method for defining chart patterns, researchers and financial analysts may define patterns in whatever way they think is suitable for their applications. Such discrepancy may lead to inconsistent and ambiguous shapes.

Although new approaches were proposed to solve the chart pattern matching problem in recent years, one of the key questions remains unsolved is how these chart patterns should be defined in an unambiguous way. To alleviate this problem, we compile formal specifications of 53 known chart patterns reported in [5]. Formal specifications have several advantages over other representations. First, the patterns defined in the formal specifications are mathematical representations. Therefore, there is no ambiguity in defining the shapes of the patterns. Second, formal specifications can be easily translated into rules that can be used to identify patterns from a given time series. Third, automated tools can be designed to verify the correctness of the formal specifications. Due to such advantages, a first-order logic representation is chosen to describe the shape and corresponding constraints of each pattern.

In financial trading, chart patterns are normally represented by either line charts or candlesticks. Therefore, formal specifications for each pattern can be viewed as the defined constraints on the relative positions of the data points (or candlesticks) within a given pattern. In addition, some of the specifications may be considered to be constraints on the temporal aspects of the patterns. Therefore, these specifications can be easily formulated into rules. Rules translated from formal specifications are not only useful for the RB method [11], but they can also be used to create standard templates in other pattern matching approaches. For example, the TB, ED or DTW methods can identify chart patterns using input templates. Normalised templates can also be used to generate synthetic data [11] [27] for experiments. To further illustrate the usefulness of formal specifications, we use rules translated from these specifications to classify patterns from a real dataset of the NYSE COMPOSITE (NYSE). To apply the RB approach, the original time series is first segmented to reduce the number of data points until the length is the same as the number of points defined in the rules. We use the perceptual important point (PIP) [7] method as a pre-processing step in the RB approach. The number of chart patterns found in the real datasets and several interesting observations are discussed in the experiment section.

The contributions of this paper can be summarized as follows:

- To the best of authors' knowledge, no one has ever defined the formal specifications of all chart patterns frequently used in the financial trading community. In this work, we endeavour to compile comprehensive formal specifications of 53 chart patterns reported in the literature. We believe that the specifications put forward in this paper can be used together as a reference model for future research in this area.

- In this paper, we argue that the formal specifications can be directly translated into rules to classify chart patterns in a real dataset from the NYSE Composite (NYSE). The experimental results show that the rules translated from the specifications can be effectively used to identify chart patterns from real datasets.

In Section 2, we review recent work on chart pattern matching. The criteria for categorizing the 53 chart patterns are extensively detailed in Section 3. The formal specifications of these patterns are given in Section 4. In Section 5, we present our experimental results. Finally, in Section 6, we conclude our paper with directions for future work.

2. Related work

Technical analysis is the study of historical stock prices to predict future price trends. In [1], Achelis introduced the basic concepts, terminology and tools of technical analysis. One of the essential tools widely used in technical analysis is stock pattern charting. In *Encyclopedia of chart patterns* [5], Bulkowski described the shape of 53 chart patterns. We define the formal specifications of the 53 chart patterns based on the descriptions given in [5]. Edwards et al. [9] also introduced technical theories, different chart patterns and other indicators in financial time series. The patterns introduced in [9] are also included in [5].

In general, pattern matching approaches require the calculation of the similarity between a chart pattern template and a query subsequence. In the TB [11], RB [11] and hybrid (HY) [27] approaches, the query subsequence and the template pattern should contain the same number of data points. In the TB approach, the similarity between the query pattern template and the segmented time series can be measured by calculating their point-to-point amplitude distances and temporal distances. In the RB approach, a set of predefined rules is used to identify the segmented time series. In the HY approach, the Spearman's correlation coefficient of the sequence is first used to rank the segmented sequence. A set of predefined rules is then used to identify the segmented sequence. When using these methods, segmentation is a necessary pre-processing step for decreasing the number of points in a time series.

The PIP method [7] is one of the most commonly used segmentation methods. In the PIP method, a set of critical points, called "PIPs", is extracted from the original subsequence. In [11], it is reported that the vertical distance PIP (PIP-VD) performs better than other variants of the same method. In [24], the effects of different segmentation methods on financial time series are evaluated. It has also been reported that the RB approach recognizes the "Head-and-Shoulders Tops" pattern with high accuracy when the PIP approach is used for segmentation. There are other segmentation methods which extract important points from the original time series, such as, the piecewise linear approximation (PLA) method [14] and the turning points (TP) method [22]. Some other segmentation methods represent the original time series in various ways. The piecewise aggregate approximation (PAA) method [15] represents a time series by the mean values of consecutive fixed-length segments. Symbolic Aggregate approXimation (SAX) [18] transforms a time series into a sequence of symbols. With SAX, a time series is firstly segmented using PAA into several parts and then each part is represented by a symbol. There are transformation techniques that represent a time series in frequency domain. Haar wavelet transform [23], which is one kind of the discrete wavelet transforms (DWT), uses Haar wavelet to represent a time series. With Haar wavelet transform technique, a time series is decomposed into two components; approximation coefficients and detail coefficients.

Segmentation is not needed in some pattern matching approaches. The ED approach uses a simple similarity measure to compare the standard template and the query subsequence. The DTW [4] approach is another popular similarity measure with many variations (e.g., DTW-D) [6]. The ED approach calculates the point-to-point Euclidean distance between two time series, whereas the DTW approach calculates the similarity between two time series that are out of phase. The DTW approach may map one point in a time series to more than one point in another time series.

Similarity measures between time series are classified into four categories in [10]. Shape-based distance compares the whole shape of the series, such as ED and DTW. Edit-based distance compares two strings by calculating the minimum number of operations needed to transform one string into another. Longest common subsequence (LCSS) metric is one of the popular measures of edit distance. Bergroth et al. [3] give a comprehensive comparison of well-known longest common subsequence algorithms. Feature-based distance measures time series based on the computation of a feature set reflecting various aspects of the series. Features can be selected by coefficients from DWT decomposition. Model-based similarity measures can also be applied in chart patterns matching in financial time series. Model-based distance works by fitting a model to different time series and then comparing the parameters of the underlying models. The hidden semi-Markov model (HSMM) has been applied in many different fields. It is commonly used in speech recognition, speech synthesis, human activity recognition and handwriting recognition. Ge et al. [12] used a HSMM to detect specific waveforms in time series. Based on [12], Kim [16] designed a HSMM with the addition of random effects to characterize waveform shape and shape variation. Traditionally, an expectation-maximisation (EM) [8] algorithm is used to estimate the parameters in an HSMM. In [16], the expectation conditional maximization either (ECME) algorithm is proposed to provide faster convergence than a standard EM procedure.

Artificial neural network (ANN) is also a popular model for pattern recognition. Zapranis and Tsinaslanidis [26] proposed a new approach based on neural networks to identify a technical pattern known as “Head-and-Shoulders”. As a class of ANN, a recurrent neural networks (RNN) [17] is a popular model for tasks that involve time series. RNNs have been widely used to solve many natural language processing problems, such as machine translation [19] and language modeling [21]. RNNs use internal memory to process time series. The hidden states of the network store information about the past elements of the sequence for later use. Long-short term memory (LSTM) [13] which is designed to avoid the long-term dependency problem is one of the training approaches of RNNs. The standard RNNs use logistic functions to calculate the hidden state, while LSTM networks compute the hidden state in a more complex way.

The first step in chart patterns matching involves defining standard templates. Some researchers use rules to describe the constraints that a chart pattern should adhere to. Andrew et al. [20] defined rules for five pairs of technical patterns (“Head-and-Shoulders”, “Inverse Head-and-Shoulders”, “Broadening Tops and Bottoms”, “Triangle Tops and Bottoms”, “Rectangle Tops and Bottoms” and “Double Tops and Bottoms”) in terms of their geometric properties with local extrema. A kernel estimator is then used to determine the extrema in the time series. Fu et al. [11] defined rule sets for “Head-and-Shoulders”, “Rounding Tops”, “Spike Tops”, “Double Tops” and “Triple Tops” using the PIP approach. In addition, they showed how to tune the thresholds of some of the rules in the rule sets. Fu et al. concluded that precision is almost unchanged for most of the chart patterns when the rules are relaxed. These rules describe the shape of chart patterns. Chart pattern language (CPL), proposed by Anand et al. [2], is a domain-specific language embedded in Haskell. CPL allows incremental composition of complex patterns from simpler patterns. The approach proposed by Anand et al. focuses on the design of a language for pattern specification, whereas our approach describes the complete formal specifications of 53 chart patterns based on their shapes, durations, start trends and breakouts. The specifications compiled in this paper can be used in the RB approach or can be translated into instructions for preparing standard templates for other pattern matching approaches.

3. Classification of chart patterns

In [5], Bulkowski provided comprehensive guidelines for identifying 53 chart patterns and how to use them to predict future price trends. We classify the chart patterns from [5] into five categories in terms of their shapes. We number the five categories from C1 to C5. Points are used to represent C1, C2 and C3, whereas candlesticks are used to represent C4 and C5.

1. **C1-Variable fluctuation patterns :** Each pattern in this category has an uncertain number of highs or lows. Therefore, the exact number of fluctuations cannot be predefined for this type of pattern. However, any instances of a pattern in this category should conform to the shape features of that particular pattern.
2. **C2-Fixed fluctuation patterns :** Each pattern in this category has a fixed number of highs or lows. Therefore the exact number of fluctuations can be predefined.
3. **C3-Curved patterns:** For patterns with curves, the shapes form various curves due to declines and increases in price. Patterns in this group usually contain a gentle rounding turn akin to a half-moon or U/inverted U. Fluctuations are formed when there is a sharp change in the price. In contrast, curves are formed when the price changes in a gentle way.
4. **C4-Candlestick patterns with spikes :** Candlesticks are used to represent patterns with spikes. A daily candlestick is a price stick that consists of the opening, highest and lowest prices in a day. Some daily candlesticks are also defined based on the closing price or both the opening and closing prices. Patterns with spikes have two remarkably longer downward and upward spikes than other downward and upward spikes around them. In these cases, the highest and lowest prices of two candlesticks are much higher and lower, respectively, than the candlesticks around them.

5. **C5-Candlestick patterns with gaps :** A gap is formed when a candlestick's highest price is lower than the lowest price of the surrounding candlesticks. A gap is also formed when a candlestick's lowest price is higher than the highest price of the surrounding candlesticks.

The price trend of the start of a chart pattern and the direction of breakout can be used to assess the reliability of a chart pattern [5]. Auxiliary lines are a kind of visual aid that is often defined in chart patterns and they are used to visualise the shape. We briefly introduce the start price trend, auxiliary line and direction of breakout in the following paragraphs.

Start price trend. The start price trend [5] of a chart pattern is the price trend leading to the formation of a chart pattern. The direction of a start price trend can be determined by searching in a backwards direction of a chart patterns x axis (time dimension). To find the initial time point of a start price trend, we begin the search at the starting point of the chart pattern and move backwards along the time dimension.

If there is an increase in price leading away from the chart pattern when moving backwards in time, we find the highest high before the price closes 20% or more below the highest high.

If there is a drop in price leading away from the chart pattern when moving backwards in time, we find the lowest low before the price closes 20% or more above the lowest low.

The "highest high" or "lowest low" marks the start of a downward or upward start price trend, respectively. As shown in Figure 1(a), the blue line is a chart pattern and Point "C" is the start of the pattern. The "highest high", Point "B", marks the start of a downward start price trend to the pattern. Point "A" is 20% or more lower than Point "B". As shown in Figure 1(b), the blue line is the pattern and Point "C" is the start of the pattern. The lowest low, Point "B", marks the start of an upward start price trend to the pattern. Point "A" is 20% or more higher than Point "B".

Auxiliary line. There are three types of auxiliary line: the trend line, the confirmation line and the neckline. Auxiliary lines are often used in chart patterns for visualization and to assess the formation of the pattern. Trend lines connect minor high or minor low points from the time series. Trend lines are often used to visualize the overall shape of chart patterns. For example, in Figure 1(c), the top trend line, which connects s_3 and s_5, and the bottom trend line, which connects s_2, s_4 and s_6, form the shape of a pattern that resembles a megaphone. Confirmation lines are mostly used to assess the formation of the pattern. When some of the points from the pattern are above or below the confirmation line, it signals the formation of the pattern. For example, Figure 1(d) depicts a pattern with a confirmation line that connects s_3 and s_5. The pattern is formed when the price of s_7 is greater than the maximum prices of s_3 and s_5. A neckline connects the low or high points situated between the shoulders of "Head-and-Shoulders" patterns (P11, P12, P23 and P24). For example, Figure 1(e) depicts a "Head-and-Shoulders Tops" pattern in which the neckline connects two points, s_3 and s_5. The necklines of "Head-and-Shoulders" patterns can also be treated as the confirmation lines.

Direction of breakout. A breakout occurs when the price of the rightmost point from the pattern pierces the auxiliary line and closes above the pattern's highest high or below the pattern's lowest low. According to [5], the price of the last point should be at least 5% higher or lower than the pattern's highest high or lowest low, otherwise the pattern cannot be considered to be reliable. Figure 1(c) depicts an upward breakout when the price of s_7 pierces the top trend line. Figure 1(f) depicts a downward breakout when the price of s_7 pierces the bottom trend line.

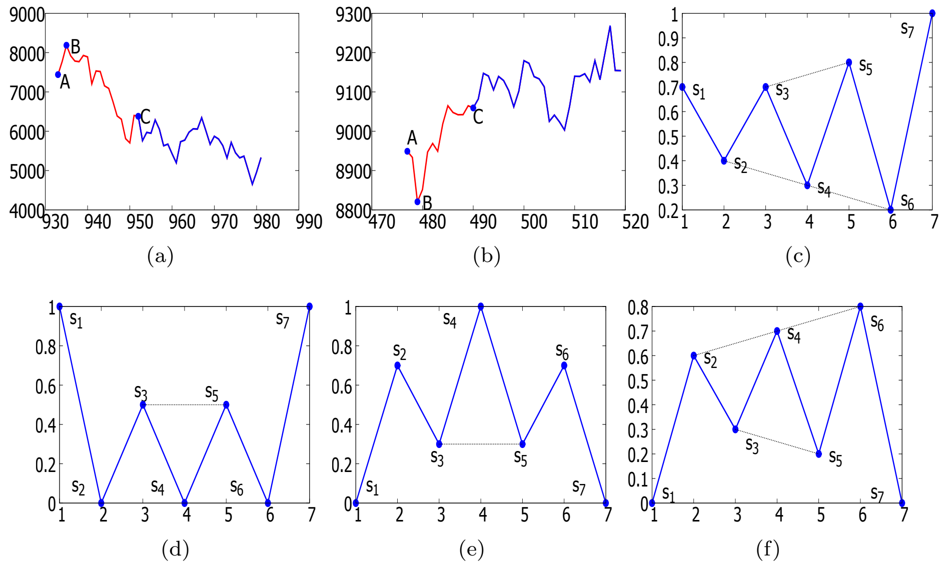


Figure 1: (a) Point "B" marks the start of a downward price trend to the start of the chart pattern Point"C". (b) Point "B" marks the start of an upward price trend to the start of the chart pattern Point "C". (c) A chart pattern with two trend lines. (d) A chart pattern with a confirmation line. (e) A "Head-and-Shoulders Tops" with a neckline. (f) A chart pattern with two trend lines.

4. Formal specifications of chart patterns

In this section, we present the formal specifications of 53 chart patterns according to the descriptions given in [5]. For the patterns in C1, C2 and C3, the specifications given in subsections 4.1, 4.2 and 4.3 can be considered to be the restrictions placed on the relationship between a series of extracted points (PIPs) from the segmented subsequence. For the patterns in C4 and C5, formal specifications describe the length of the spikes or the relationship between the highest or lowest prices among spikes. Therefore, the specifications given in 4.4 and 4.5 can be considered to be restrictions on the relationship between the highest price and the lowest price of the candlesticks.

We use the PIP method to extract a specified number of PIPs (salient points) from a time series. These extracted points are used for pattern matching in the experiment section. We can also use other segmentation methods, such as piecewise aggregate approximation (PAA) [15], piecewise linear approximation (PLA) [14] or the turning points (TP) method [22].

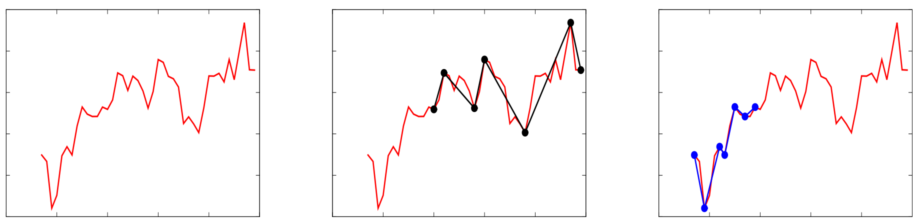
Definitions and functions. In the following paragraphs, we describe the definitions and functions that specify the contours, start trends and durations of chart patterns.

Definition 1 (Time series) A time series $T = \{(x_1, y_1), (x_2, y_2), ..., (x_n, y_n)\}$ consists of a sequence of points. x_i is the time coordinate of a point and y_i is the price of the point.

Definition 2 (Sequence of salient points to be classified) $S = \{(x_1, y_1), (x_2, y_2), ..., (x_n, y_n)\}$ is a subsequence of T and includes a sequence of salient points. S can be considered to be a query subsequence for classification. x_i and y_i are the time coordinate and price of the salient point, s_i. The identifier of the first salient point is 1 and the identifier of the last salient point is n. Without loss of generality, s_i is used to denote (x_i, y_i). $S[s_i, s_j]$ is a subsequence from s_i to s_j of S.

Definition 3 (The head sequence of a start trend) A sequence of salient points $R = \{(x_1, y_1), (x_2, y_2), ..., (x_n, y_n)\}$ is the head sequence of a start trend when R is followed by another sequence, S.

Figure 2(a) shows a time series, T. Figure 2(b) shows a sequence of salient points, S, (black) that will be classified as a chart pattern or not. We use the PIP method to extract the sequence of salient points, S, from the time series, T. Figure 2(c) shows a sequence of salient points, R (blue), that can be used to describe the direction of the start price trend preceding the sequence, S. In this paper, R is a sequence of local maximum and minimum points extracted from time series T.



(a) (b) (c)

Figure 2: (a) A time series, T. (b) A sequence of salient points, S (black), on the time series, T (red). (c) A head sequence of a start trend, R (blue), on time series T (red).

Function 1 $len(S)$ returns the length or the number of salient points of the sequence, S.

Function 2 $get\_time(s_i)$ returns the time stamp of point s_i. $get\_time((x_i, y_i)) = x_i$

Function 3 $get\_price(s_i)$ returns the price of point s_i. $get\_price((x_i, y_i)) = y_i$

Function 4 $get\_id(s_i)$ returns the identifier of point s_i. $get\_id((x_i, y_i)) = i$.

Function 5 $odd(s_i)$ returns true if i is an odd number.

Function 6 $even(s_i)$ returns true if i is an even number.

Function 7 (Odd-numbered sequence of salient points) $seq\_odd(S)$ is a sequence of odd-numbered salient points (except the first salient point and the last salient point) of S.
$seq\_odd(S) = \{s_i | s_i \in S \wedge odd(s_i) \wedge i \neq 1 \wedge i \neq n\}$.

Function 8 (Even-numbered sequence of salient points) $seq\_even(S)$ is a sequence of even-numbered salient points (except the first salient point and the last salient point) of S.
$seq\_even(S) = \{s_i | s_i \in S \wedge even(s_i) \wedge i \neq 1 \wedge i \neq n\}$.

Function 9 $increase(S)$ returns true if the prices of the salient points of the smaller identifier are less than those of the greater identifier in sequence S.
$increase(S) \equiv \forall s_i \in S, s_j \in S \wedge i<j \Rightarrow get\_price(s_i)<get\_price(s_j)$.

Function 10 $decrease(S)$ returns true if the prices of the salient points of the smaller identifier are greater than those of the greater identifier in sequence S.

$decrease(S) \equiv \forall s_i \in S, s_j \in S \land i<j \Rightarrow get\_price(s_i)>get\_price(s_j)$.

Definition 4 (Minor low) $minor\_low(s_i)$ denotes a salient point, s_i, the price of which is less than that of the previous salient point and the next salient point.

$minor\_low(s_i) \equiv get\_price(s_i)<get\_price(s_{i+1}) \land get\_price(s_i)<get\_price(s_{i-1})$.

Definition 5 (Minor high) $minor\_high(s_i)$ denotes a salient point, s_i, the price of which is greater than that of the previous salient point and the next salient point.

$minor\_high(s_i) \equiv get\_price(s_i)>get\_price(s_{i+1}) \land get\_price(s_i)>get\_price(s_{i-1})$.

Function 11 $vd(s_i, s_k, s_j)$ returns the vertical distance from the salient point s_k to the line connecting two salient points, s_i and s_j.

$vd(s_i, s_k, s_j) = \left| get\_price(s_i) + \frac{(get\_price(s_j)-get\_price(s_i))\times(get\_time(s_k)-get\_time(s_i))}{(get\_time(s_j)-get\_time(s_i))} - get\_price(s_k) \right|$

Function 12 $slope(s_i, s_j)$ returns the slope of two points.

$slope(s_i, s_j) = \frac{get\_price(s_i)-get\_price(s_j)}{get\_time(s_i)-get\_time(sj)}$

Function 13 $argmax(S)$ returns the identifier of the salient point of the highest price in sequence S.

$argmax(S) = \{i|\forall s_j \in S \land i \neq j \Rightarrow get\_price(s_j) < get\_price(s_i)\}$

Function 14 $argmin(S)$ returns the identifier of the salient point of the lowest price in sequence S.

$argmin(S) = \{i|\forall s_j \in S \land i \neq j \Rightarrow get\_price(s_j) > get\_price(s_i)\}$

Function 15 $dif\_price(s_i, s_j)$ returns the price difference of two salient points.

$dif\_price(s_i, s_j) = |get\_price(s_i) - get\_price(s_j)|$

Function 16 $dif\_time(s_i, s_j)$ returns the time difference of two salient points.

$dif\_time(s_i, s_j) = |get\_time(s_i) - get\_time(s_j)|$

Function 17 $similar\_price(s_i, s_j)$ returns true if the price difference of two salient points is less than or equal to 10% of the difference of the highest price and the lowest price of sequence S.

$similar\_price(s_i, s_j) \equiv dif\_price(s_i, s_j) \leq 0.1 \times dif\_price(s_{argmax(S)}, s_{argmin(S)})$

Function 18 $uniform\_price(S)$ returns true if the prices of the salient points in sequence S are at a similar price level.

$uniform\_price(S) \equiv \forall s_i \in S, s_j \in S \land j \neq i \Rightarrow similar\_price(s_i, s_j)$

Function 19 $sym\_similar\_price(s_i, S)$ returns true if the prices of the points in the sequence S are symmetrical at the same price level around the point s_i. For example, given a sequence $S = (s_1, s_2, s_3, s_4, s_5, s_6, s_7)$, $sym\_similar\_price(s_4, S)$ returns true if two pairs of points (s_3 and s_5 , s_2 and s_6) are at similar price levels, respectively.

$sym\_similar\_price(s_i, S) \equiv \forall s_j \in S, \exists s_k \in S \land |k-i| = |j-i| \land j \neq 1 \land j \neq n \Rightarrow similar\_price(s_j, s_k)$

Function 20 $similar\_time\_interval(s_i, s_j, s_k, s_g)$ returns true if the difference of the two time differences (the time difference between s_i and s_j and the time difference between s_k and s_g) is less than or equal to 10% of the difference of the maximum time and the minimum time of sequence S.

$similar\_time\_interval(s_i, s_j, s_k, s_g) \equiv |dif\_time(s_i, s_j) - dif\_time(s_k, s_g)| \leq 0.1 \times dif\_time(s_n, s_1)$

Function 21 $similar\_price\_interval(s_i, s_j, s_k, s_g)$ returns true if the difference of the two price differences (the price difference between s_i and s_j and the price difference between s_k and s_g) is less than or equal to 10% of the difference of the highest price and lowest price of sequence S.

$similar\_price\_interval(s_i, s_j, s_k, s_g) \equiv |dif\_price(s_i, s_j) - dif\_price(s_k, s_g)| \leq 0.1 \times dif\_price(s_{argmax(S)}, s_{argmin(S)})$

Function 22 $dif\_price\_range(s_i, s_j, \beta, \gamma)$ returns true if the difference of the prices of two points, s_i and s_j, is greater than or equal to β times the difference of the highest price and the lowest price of sequence S and less than or equal to γ times the difference of the highest price and the lowest price of sequence S.

$dif\_price\_range(s_i, s_j, a, b) \equiv \beta \times dif\_price(s_{argmax(S)}, s_{argmin(S)}) \leq dif\_price(s_i, s_j)$
$\leq \gamma \times dif\_price(s_{argmax(S)}, s_{argmin(S)}), \beta \in (0, 1) \wedge \gamma \in (0, 1)$.

Function 23 $sym\_similar\_time\_interval(s_i, S)$ returns true if the time intervals in sequence S are symmetrically similar around the salient point, s_i. For example, given a sequence $S = (s_1, s_2, s_3, s_4, s_5, s_6, s_7)$, $sym\_similar\_time\_interval(s_4, S)$ returns true if two pairs of time differences (the time difference between s_3 and s_4 and the time difference between s_5 and s_4, the time difference between s_2 and s_4 and the time difference between s_6 and s_4) are similar.

$sym\_similar\_time\_interval(s_i, S) \equiv \forall s_j \in S, \exists s_k \in S \wedge |k - i| = |j - i| \wedge j \neq 1 \wedge j \neq n \Rightarrow similar\_time\_interval(s_i, s_j, s_i, s_k)$

Function 24 $odd\_head(s_i, S)$ retrieves a sequence of points that have similar prices to point s_i from sequence S.

$odd\_head(s_i, S) = \{s_j | s_j \in S \wedge j \neq 2 \wedge j \neq n - 1 \Rightarrow similar\_price(s_i, s_j)\}$

Function 25 $even\_head(s_i, s_j, S)$ retrieves a sequence of points that have similar prices to points s_i and s_j from sequence S.

$even\_head(s_i, s_j, S) = \{s_k | s_k \in S \wedge k \neq 2 \wedge k \neq n-1 \Rightarrow similar\_price(s_i, s_k) \wedge similar\_price(s_j, s_k)\}$

Function 24 and Function 25 are used to describe definitions of “Head-and-Shoulders Tops, Complex” (P11) and “Head-and-Shoulders Bottoms, Complex” (P12). “Head-and-Shoulders Tops, Complex” or “Head-and-Shoulders Bottoms, Complex” either has an odd number of heads or has an even number of heads. Function 24 is used to describe the pattern of odd-numbered heads, whereas Function 25 is used to describe the pattern of even-numbered heads.

Function 26 $up\_trend(R, S)$ returns true if the start price trend of sequence S is upward. If the price climbs leading away from S when moving backwards in time, find the highest high before the price closes 20% or more below and before the highest high. The highest high marks the start of an upward start price trend. A detailed discussion about the upward start price trend is given in subsection 3. R describes the start price trend as shown in Definition 3.

$up\_trend(R, S) \equiv \exists r_i \in R, r_{i-1} \in R \wedge minor\_low(r_i) \wedge minor\_high(r_{i-1}) \Rightarrow dif\_price(r_i, r_{i-1}) \geq 0.2 \times dif\_price(s_{argmax(R,S)}, s_{argmin(R,S)})$

Function 27 $down\_trend(R, S)$ returns true if the start price trend of the sequence S is downward. If the price drops leading away from S when moving backwards in time, find the lowest low before the price closes 20% or more above and before the lowest low. The lowest low marks the start of a downward start price trend. A detailed discussion about the downward start price trend is given in subsection 3. R describes the start price trend (see Definition 3).

$down\_trend(R,S) \equiv \exists r_i \in R, r_{i-1} \in R \wedge minor\_high(r_i) \wedge minor\_low(r_{i-1}) \Rightarrow dif\_price(r_i, r_{i-1}) \geq 0.2 \times dif\_price(s_{argmax(R,S)}, s_{argmin(R,S)})$

In some cases, when the first point of S is also the first point of R, both Function 26 and Function 27 return false. In these cases, we cannot decide the start price trend of S. When we encounter this case in the experiment, the start price trend of S is considered to be downward.

4.1. C1-Variable fluctuation patterns

Variable fluctuation patterns (C1) have an uncertain number of minor highs and minor lows; therefore, we cannot define their shapes by a fixed number of salient points. We use a variable, n, to represent that the patterns have an uncertain number of salient points. In patterns with fluctuations, as the term "fluctuations" suggests, all of the patterns of C1 are formed by points that are either minor lows or minor highs, except for the first point and the last point.

For three pairs of patterns ("Broadening Bottoms" (P1) and "Broadening Tops" (P4), "Diamond Bottoms" (P7) and "Diamond Tops" (P8) and "Rectangle Bottoms" (P14) and "Rectangle Tops" (P15)), the difference between bottoms and tops is the direction of the start price trend. "Flags" (P9) has the same shape as "Rectangle Bottoms" and "Rectangle Tops" (P14 and P15), except the duration of "Flags" is shorter and "Flags" has no restriction on the direction of a start price trend. "Flags, High and Tight" (P10) has the same shape as "Rectangle Bottoms" and "Rectangle Tops" (P14 and P15), except the duration of "Flags, High and Tight" is shorter. "Pennants" (P13) has the same shape as "Symmetrical Triangles" (P18), "Wedges, Falling" (P19) and "Wedges, Rising" (P20), except the duration of "Pennants" is shorter.

Pattern P1 (Broadening Bottoms).

Description. The direction of the pattern's start price trend is downward. The shape resembles a megaphone with higher highs and lower lows. "Broadening Bottoms" has two trend lines. The top line slopes up and the bottom line slopes down. At least two minor lows and two minor highs touch the trend lines.

There are four kinds of "Broadening Bottoms". We number the four variations as P1-1, P1-2, P1-3 and P1-4. Due to limited space, only one variation is shown. However, the other variations have the same shape features as the first variation. P1-1 is a variation of "Broadening Bottoms" in which the second salient point is a minor high and the last-but-one salient point is a minor low. P1-2 is a variation in which the second salient point is a minor high and the last-but-one salient point is a minor high. P1-3 is a variation in which the second salient point is a minor low and the last-but-one salient point is a minor low. P1-4 is a variation in which the second salient point is a minor low and the last-but-one salient point is a minor high. The four variations of each pattern in C1 (except for "Head-and-Shoulders Bottoms, Complex" (P11) and "Head-and-Shoulders Tops, Complex" (P12)) have the same features of the second and last-but-one salient points as the four variations of "Broadening Bottoms". We use the direction of the start price trend to distinguish between "Broadening Bottoms" and "Broadening Tops" (P4).

P1-1: $even(s_n) \wedge (len(S) \geq 6) \wedge down\_trend(R, S) \wedge increase(seq\_even(S)) \wedge decrease(seq\_odd(S))$
$(\forall s_i \in seq\_even(S) \Rightarrow minor\_high(s_i)) \wedge (\forall s_i \in seq\_odd(S) \Rightarrow minor\_low(s_i))$

<u>Pattern</u> *P2 (Broadening Formations, Right-Angled and Ascending).*

Description. The shape resembles a megaphone with higher highs and similar lows. It has two trend lines. The top line slopes up and the bottom line is horizontal. At least two minor lows and two minor highs touch the trend lines. There are four kinds of "Broadening Formations, Right-Angled and Ascending". We number the four variations as P2-1, P2-2, P2-3 and P2-4.

P2-1: $even(s_n) \wedge (len(S) \geq 6) \wedge increase(seq\_even(S)) \wedge uniform\_price(seq\_odd(S))$
$(\forall s_i \in seq\_even(S) \Rightarrow minor\_high(s_i)) \wedge (\forall s_i \in seq\_odd(S) \Rightarrow minor\_low(s_i))$

<u>Pattern</u> *P3 (Broadening Formations, Right-Angled and Descending).*

Description. The shape resembles a megaphone with similar highs and lower lows. It has two trend lines. The top line is horizontal and the bottom line slopes down. At least two minor lows and two minor highs touch the trend lines. There are four kinds of "Broadening Formations, Right-Angled and Descending". We number the four variations as P3-1, P3-2, P3-3 and P3-4.

P3-1: $even(s_n) \wedge (len(S) \geq 6) \wedge decrease(seq\_odd(S)) \wedge unifrom\_price(seq\_even(S))$
$(\forall s_i \in seq\_even(S) \Rightarrow minor\_high(s_i)) \wedge (\forall s_i \in seq\_odd(S) \Rightarrow minor\_low(s_i))$

<u>Pattern</u> *P4 (Broadening Tops).*

Description. This pattern's start price trend is upward. The shape resembles a megaphone with higher highs and lower lows. It has two trend lines. The top line slopes up and the bottom line slopes down. At least two minor lows and two minor highs touch the trend lines. The shape of "Broadening Tops" is actually the same as "Broadening Bottoms" (P1), the difference between them is the direction of the start price trend. "Broadening Tops" has the same definition as "Broadening Bottoms", except the trend at the start of the pattern is upward.

<u>Pattern</u> *P5 (Broadening Wedges, Ascending).*

Description. The shape resembles a megaphone with higher highs and higher lows. It has two trend lines. Both of the trend lines slope up with the top line being steeper than the bottom line. The top up-sloping trend line is steeper than the bottom up-sloping trend line corresponding to the definition that the slope of the top trend line is greater than the slope of the bottom trend line. At least three minor lows and three minor highs touch the trend lines. There are four kind-s of "Broadening Wedges, Ascending". We number the four variations as P5-1, P5-2, P5-3 and P5-4.

P5-1: $even(s_n) \wedge (len(S) \geq 8) \wedge increase(seq\_even(S)) \wedge increase(seq\_odd(S))$
$(\forall s_i \in seq\_even(S) \Rightarrow minor\_high(s_i)) \wedge (\forall s_i \in seq\_odd(S) \Rightarrow minor\_low(s_i))$
$slope(s_2, s_{n-2}) > slope(s_3, s_{n-1})$

Pattern P6 (Broadening Wedges, Descending).

Description. The shape resembles a megaphone with lower highs and lower lows. It has two trend lines. Both of the trend lines slope down and the bottom trend line is steeper than the top trend line. The bottom down-sloping trend line is steeper than the top down-sloping trend line corresponding to the definition that the slope of the top trend line is greater than the slope of the bottom trend line. At least two minor lows and two minor highs touch the trend lines. There are four kinds of "Broadening Wedges, Descending". We number the four variations as P6-1, P6-2, P6-3 and P6-4.

P6-1: $even(s_n) \wedge (len(S) \geq 6) \wedge decrease(seq\_even(S)) \wedge decrease(seq\_odd(S))$
$(\forall s_i \in seq\_even(S) \Rightarrow minor\_high(s_i)) \wedge (\forall s_i \in seq\_odd(S) \Rightarrow minor\_low(s_i))$
$slope(s_2, s_{n-2}) > slope(s_3, s_{n-1})$

Pattern P7 (Diamond Bottoms).

Description. This pattern's start price trend is downward. "Diamond Bottoms" has higher highs and lower lows in the first part, followed by lower highs and higher lows in the second part. "Diamond Bottoms" does not need to be symmetrical. Four trend lines form a diamond. There are four kinds of "Diamond Bottoms". We number the four variations as P7-1, P7-2, P7-3 and P7-4. We use the direction of start price trend to distinguish between "Diamond Bottoms" and "Diamond Tops" (P8).

P7-1: $even(s_n) \wedge (len(S) \geq 8) \wedge even(s_{argmax(S)}) \wedge odd(s_{argmin(S)}) \wedge down\_trend(R, S)$
$(\forall s_i \in seq\_even(S) \Rightarrow minor\_high(s_i)) \wedge (\forall s_i \in seq\_odd(S) \Rightarrow minor\_low(s_i))$
$increase(seq\_even(S)[s_2, s_{argmax(S)}]) \wedge decrease(seq\_even(S)[s_{argemax(S)}, s_{n-2}])$
$decrease(seq\_odd(S)[s_3, s_{argmin(S)}]) \wedge increase(seq\_odd(S)[s_{argmin(S)}, s_{n-1}])$

Pattern P8 (Diamond Tops).

Description. This pattern's start price trend is upward. "Diamond Tops" has higher highs and lower lows in the first part, followed by lower highs and higher lows in the second part. "Diamond Tops" does not need to be symmetrical. Four trend lines form a diamond. The difference between a "Diamond Bottoms" (P7) and a "Diamond Tops" is the start price trend. "Diamond Tops" has the same definition as "Diamond Bottom", except its start price trend is upward.

Pattern P9 (Flags).

Description. "Flags" can be "Rectangle Bottoms" or "Rectangle Tops" with durations shorter than 3 weeks. The two trend lines of "Flags" should be parallel. "Flags" has the same definition as "Rectangle Bottoms" or "Rectangle Tops", except its duration is shorter than 3 weeks.

<u>Pattern</u> P10 (Flags, High and Tight).

Description. The chart pattern "Flags, High and Tight" is a special case of the "Flags" chart pattern. The difference between these two chart patterns is their start price trends. We use the same definition as "Flags" with the added constraint that the start price trend of "Flags, High and Tight" is upward.

<u>Pattern</u> P11 (Head-and-Shoulders Bottoms, Complex).

Description. "Head-and-Shoulders Bottoms, Complex" has multiple shoulders, multiple heads or (rarely) both. The head is lower than the shoulders. The price level of the shoulders and time distance from the shoulder to head is approximately the same on either side of the head. In addition, the shoulders appear to be the same shape, that is, narrow or wide shoulders on the left mirror the right. A neckline connects both the highest rise on the left and on the right. Many formations have near-horizontal necklines. The pattern has an upward breakout of the neckline.

"Head-and-Shoulders Bottoms, Complex" has two variations: P11-1 and P11-2. P11-1 is a variation of "Head-and-Shoulders Bottoms, Complex" in which the number of heads is an odd number, whereas P11-2 is a variation of "Head-and-Shoulders Bottoms, Complex" in which the number of heads is an even number.

P11-1: The number of heads is an odd number

$$odd(s_n) \wedge even(s_{\frac{n+1}{2}}) \wedge (len(S) \geq 11) \wedge get\_price(s_n) \geq get\_price(s_{n-2})$$
$$(\forall s_i \in seq\_even(S) \Rightarrow minor\_low(s_i)) \wedge (foralls_i \in seq\_odd(S) \Rightarrow minor\_high(s_i))$$
$$\exists s_i \in odd\_head(s_{\frac{n+1}{2}}, seq\_even(S)) \Rightarrow i = argmin(S)$$
$$sym\_similar\_price(s_{\frac{n+1}{2}}, S) \wedge sym\_similar\_time\_interval(s_{\frac{n+1}{2}}, seq\_even(S))$$

<u>Pattern</u> P12 (Head-and-Shoulders Tops, Complex).

Description. "Head-and-Shoulders Tops, Complex" has multiple shoulders or, more rarely, two heads. The head is higher than the shoulders. The shoulders mirror themselves about the head. A neckline connects the lowest left shoulder trough with the lowest right shoulder trough. The pattern has a downward breakout of the neckline.

There are two variations of "Head-and-Shoulders Tops, Complex": P12-1 and P12-2. P12-1 is a variation of the "Head-and-Shoulders Tops, Complex" in which the number of heads is an odd number, whereas P12-2 is a variation of the "Head-and-Shoulders Tops, Complex" in which the number of heads is an even number.

P12-1: The number of heads is an odd number

$$odd(s_n) \wedge even(s_{\frac{n+1}{2}}) \wedge (len(S) \geq 11) \wedge get\_price(s_n) \leq get\_price(s_{n-2})$$
$$(\forall s_i \in seq\_even(S) \Rightarrow minor\_high(s_i)) \wedge (\forall s_i \in seq\_odd(S) \Rightarrow minor\_low(s_i))$$
$$\exists s_i \in odd\_head(s_{\frac{n+1}{2}}, seq\_even(S)) \Rightarrow i = argmax(S)$$
$$sym\_similar\_price(s_{\frac{n+1}{2}}, S) \wedge sym\_similar\_time\_interval(s_{\frac{n+1}{2}}, seq\_even(S))$$

<u>Pattern</u> *P13 (Pennants).*

Description. "Pennants" can be "Triangle, Symmetrical" or "Wedges, (Rising or Falling)", except its duration is 3 weeks maximum. "Pennants" have the same definition as "Triangle, Symmetrical" and "Wedges, (Rising or Falling)", except its duration is shorter than 3 weeks.

<u>Pattern</u> *P14 (Rectangle Bottoms).*

Description. The start price trend is downward and then the pattern oscillates between two horizontal trend lines. There are at least two touches of each trend line. A "Rectangle Bottoms" or "Rectangle Tops" that is shorter than 3 weeks is a "Flags", so the duration of "Rectangle Bottoms" should be longer than 3 weeks. There are four kinds of "Rectangle Bottoms". We number the four variations as P14-1, P14-2, P14-3 and P14-4.

P14-1: $even(s_n) \wedge (len(S) \geq 6) \wedge down\_trend(R, S) \wedge uniform\_price(seq\_even(S)) \wedge uniform\_price(seq\_odd(S))$
$$(\forall s_i \in seq\_even(S) \Rightarrow minor\_high(s_i)) \wedge (\forall s_i \in seq\_odd(S) \Rightarrow minor\_low(s_i))$$

<u>Pattern</u> *P15 (Rectangle Tops).*

Description. The start price trend is upward and then the pattern oscillates between two horizontal trend lines. There are at least two touches of each trend line. The difference between "Rectangle Bottoms" and "Rectangle Tops" is the start price trend. "Rectangle Tops" has the same definition as "Rectangle Bottoms", except its start price trend is upward.

<u>Pattern</u> *P16 (Triangles, Ascending).*

Description. A horizontal top trend line and an up-sloping bottom trend line form "Triangles, Ascending". There are at least two high touches on the top horizontal line and at least two low touches on the up-sloping bottom trend line. There are four kinds of "Triangles, Ascending". We number the four variations as P16-1, P16-2, P16-3 and P16-4.

P16-1: $even(s_n) \wedge (len(S) \geq 6) \wedge unifrom\_price(seq\_even(S)) \wedge increase(seq\_odd(S))$
$$(\forall s_i \in seq\_even(S) \Rightarrow minor\_high(s_i)) \wedge (\forall s_i \in seq\_odd(S) \Rightarrow minor\_low(s_i))$$

<u>Pattern</u> *P17 (Triangles, Descending).*

Description. "Triangles, Descending" has a horizontal bottom trend line and a down-sloping top trend line. At least two minor lows and two minor highs touch the trend lines. There are four kinds of "Triangles, Descending". We number the four variations as P17-1, P17-2, P17-3 and P17-4.

P17-1: $even(s_n) \wedge (len(S) \geq 6) \wedge unifrom\_price(seq\_odd(S)) \wedge decrease(seq\_even(S))$
$(\forall s_i \in seq\_even(S) \Rightarrow minor\_high(s_i)) \wedge (\forall s_i \in seq\_odd(S) \Rightarrow minor\_low(s_i))$

<u>Pattern</u> *P18 (Triangles, Symmetrical).*

Description. The price forms lower highs and higher lows following two sloping trend lines that eventually intersect. It has two trend lines. One slopes downward from the top and the other slopes upward from the bottom such that the two lines join at the apex. At least two minor highs and minor lows touch the trend lines. The minimum duration of "Triangles, Symmetrical" is 3 weeks (approximately 21 days). There are four kinds of "Triangles, Symmetrical". We number the four variations as P18-1, P18-2, P18-3 and P18-4.

P18-1: $dif\_time(s_1, s_n){>}21 \wedge even(s_n) \wedge (len(S) \geq 6) \wedge decrease(seq\_even(S)) \wedge increase(seq\_odd(S))$
$(\forall s_i \in seq\_even(S) \Rightarrow minor\_high(s_i)) \wedge (\forall s_i \in seq\_odd(S) \Rightarrow minor\_low(s_i))$

<u>Pattern</u> *P19 (Wedges, Falling).*

Description. "Wedge Fallings" has two trend lines. The trend lines must both slope downward and eventually intersect with the top trend line, which has a steeper slope than the bottom trend line. The top down-sloping trend line is steeper than the bottom down-sloping trend line corresponding to the definition that the slope of the bottom trend line is greater than the slope of the top trend line. "Wedge Fallings" has at least five touches (three on one side and two on the other). The minimum duration of the pattern is 3 weeks. The duration rarely exceeds 4 months.

There are four kinds of "Wedge Fallings". We number the four variations as P19-1, P19-2, P19-3 and P19-4.

P19-1: $(dif\_time(s_1, s_n){>}21) \wedge even(s_n) \wedge (len(S) \geq 6)$
$(\forall s_i \in seq\_even(S) \Rightarrow minor\_high(s_i)) \wedge (\forall s_i \in seq\_odd(S) \Rightarrow minor\_low(s_i))$
$decrease(seq\_even(S)) \wedge decrease(seq\_odd(S)) \wedge (slope(s_2, s_{n-2}){<}slope(s_3, s_{n-1}))$

<u>Pattern</u> *P20 (Wedges, Rising).*

Description. An upward price spiral is bounded by two intersecting, up-sloping trend lines. Both trend lines must have upward slopes and eventually intersect with the bottom trend line producing a steeper slope than that at the top. The bottom up-sloping trend line is steeper than the top up-sloping trend line corresponding to the definition that the slope of the bottom trend line is greater than the slope of the top trend line. A "Wedges, Rising" pattern should have at least five touches (three on one side and two on the other). The minimum duration of the pattern is 3 weeks. The duration rarely exceeds 3 or 4 months. There are four kinds of "Wedges, Rising". We number the four variations as P20-1, P20-2, P20-3 and P20-4.

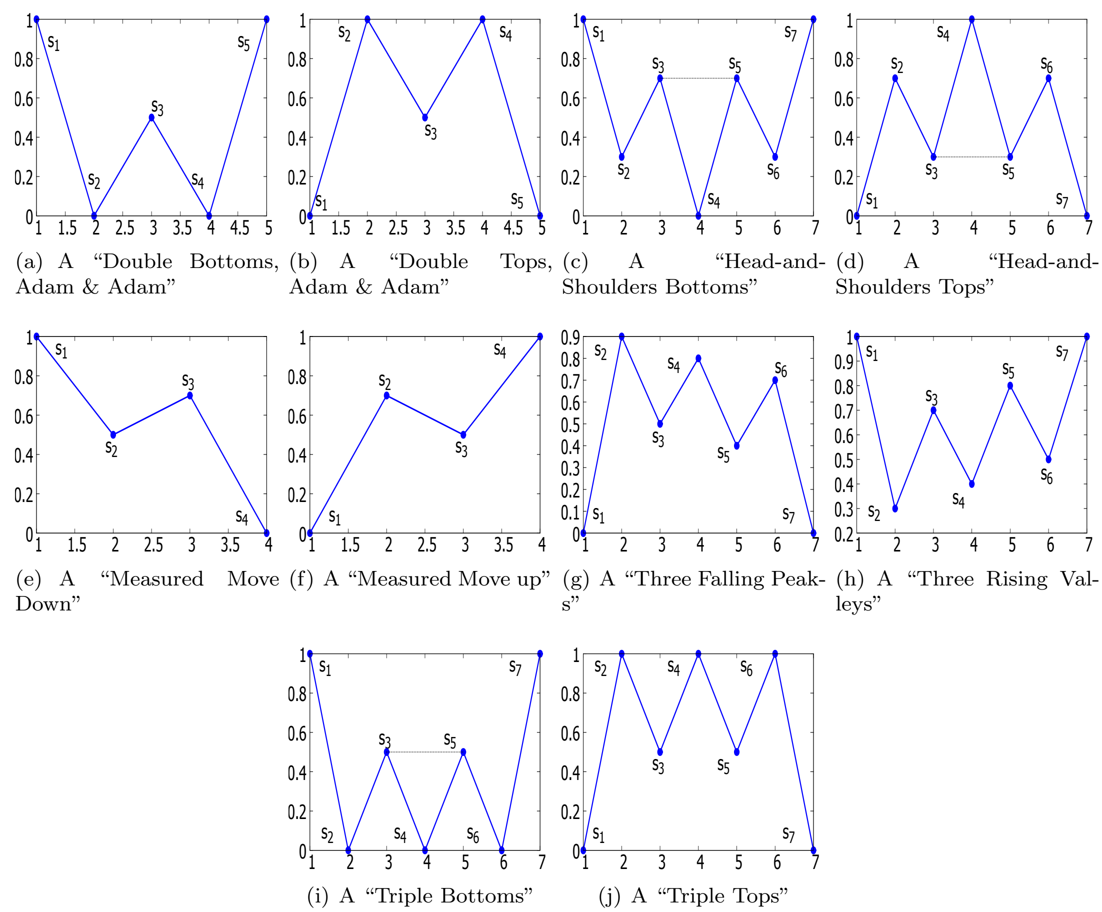
P20-1: $(dif\_time(s_1, s_n) > 21) \wedge even(s_n) \wedge len(S) \geq 6$

$$(\forall s_i \in seq\_even(S) \Rightarrow minor\_high(s_i)) \wedge (\forall s_i \in seq\_odd(S) \Rightarrow minor\_low(s_i))$$

$$increase(seq\_even(S)) \wedge increase(seq\_odd(S)) \wedge (slope(s_2, s_{n-2}) < slope(s_3, s_{n-1}))$$

4.2. C2-Fixed fluctuation patterns

Fixed fluctuation patterns (C2) have a certain number of minor highs and minor lows. We can define them by a fixed number of points.



(a) A "Double Bottoms, Adam & Adam" (b) A "Double Tops, Adam & Adam" (c) A "Head-and-Shoulders Bottoms" (d) A "Head-and-Shoulders Tops" (e) A "Measured Move Down" (f) A "Measured Move up" (g) A "Three Falling Peaks" (h) A "Three Rising Valleys" (i) A "Triple Bottoms" (j) A "Triple Tops"

Figure 3: Example patterns in C2

<u>Pattern</u> P21 (Double Bottoms, Adam & Adam).

Description. This pattern has two narrow, V-shaped bottoms that have similar prices. A rise between bottoms from the lowest valley should be at least 10%. The confirmation line is a horizontal line drawn by the highest high (s_3 in Figure 3(a)) between the two bottoms. The price must close above the confirmation point (s_3 in Figure 3(a)). The separation between the two bottoms is a few

(3 to 6) weeks.

$$\textbf{P21:}\ (S=\{s_1,s_2,s_3,s_4,s_5\}) \wedge (len(S)=5) \wedge (21<dif\_time(s_2,s_4)<42)$$
$$minor\_low(s_2) \wedge minor\_low(s_4) \wedge (get\_price(s_5)>get\_price(s_3))$$
$$similar\_price(s_2,s_4) \wedge \neg similar\_price(s_3,min(s_2,s_4))$$

Pattern P22 (Double Tops, Adam & Adam).

Description. The price trend at the start of the formation is upward. The pattern has two narrow, V-shaped tops that have similar prices. A larger dip between the two tops performs better. The confirmation line is a horizontal line drawn by the lowest low (s_3 in Figure 3(b)) between the two tops. The price must close below the confirmation point (s_3 in Figure 3(b)). The separation between the two tops is a few (5 to 7) weeks.

$$\textbf{P22:}\ (S=\{s_1,s_2,s_3,s_4,s_5\}) \wedge (len(S)=5) \wedge (35<dif\_time(s_2,s_4)<49)$$
$$minor\_high(s_2) \wedge minor\_high(s_4) \wedge (get\_price(s_5)<get\_price(s_3))$$
$$similar\_price(s_2,s_4) \wedge \neg similar\_price(s_3,min(s_2,s_4))$$

Pattern P23 (Head-and-Shoulders Bottoms).

Description. A three-trough formation with the centre trough below the other two. The right shoulder declines to approximately the price level of the left shoulder and the distances of both shoulders from the head are similar. The neckline (the line connecting s_3 and s_5 in Figure 3(c)) is a line that connects the rise between the two shoulders. A neckline pierce signals an upward breakout.

$$\textbf{P23:}\ (S=\{s_1,s_2,s_3,s_4,s_5,s_6,s_7\}) \wedge (len(S)=7) \wedge (get\_price(s_7)\geq get\_price(s_5))$$
$$minor\_low(s_2) \wedge minor\_low(s_4) \wedge minor\_low(s_6) \wedge minor\_high(s_3) \wedge minor\_high(s_5)$$
$$similar\_price(s_2,s_6) \wedge similar\_price(s_3,s_5) \wedge similar\_time\_interval(s_4,s_2,s_6,s_4)$$

Pattern P24 (Head-and-Shoulders Tops).

Description. "Head-and-Shoulders Tops" has a three-peak formation with a centre peak that is taller than the others. The two shoulders appear at approximately the same price level and the distance from the shoulders to the head is approximately the same. The neckline (the line connecting s_3 and s_5 in Figure 3(d)) connects the two lows between the three peaks. A downward breakout occurs when price closes below the neckline.

$$\textbf{P24:}\ (S=\{s_1,s_2,s_3,s_4,s_5,s_6,s_7\}) \wedge (len(S)=7) \wedge (get\_price(s_7)\leq get\_price(s_5))$$
$$minor\_high(s_2) \wedge minor\_high(s_4) \wedge minor\_high(s_6) \wedge minor\_low(s_3) \wedge minor\_low(s_5)$$
$$similar\_price(s_2,s_6) \wedge similar\_price(s_3,s_5) \wedge similar\_time\_interval(s_4,s_2,s_6,s_4)$$

Pattern P25 (Measured Move Down).

Description. The price moves down, retraces and then moves down again. At the first leg, the price declines rapidly in a straight line. The price then usually moves up to rise and recover from 38% to 62% of the decline or move horizontally before resuming the downward trend. Finally, the downward movement resumes. The second leg decline approximates the price decline set by the first leg and the time it took to accomplish it. See Figure 3(e).

P25: $(S = \{s_1, s_2, s_3, s_4\}) \land (len(S) = 4) \land minor\_low(s_2)$
$(get\_price(s_3) > get\_price(s_4)) \land dif\_price\_range(s_2, s_3, 0.38, 0.62)$
$similar\_price\_interval(s_1, s_2, s_3, s_4) \land similar\_time\_interval(s_1, s_2, s_3, s_4)$

<u>Pattern</u> P26 (Measured Move Up).

Description. The price moves up, retraces and then moves up again. The first leg follows a straight course upward. Then, the price usually declines to between 40% and 60% of the first leg move. The second leg rise is usually proportional to the first leg rise. See Figure 3(f).

P26: $(S = \{s_1, s_2, s_3, s_4\}) \land (len(S) = 4) \land minor\_high(s_2)$
$(get\_price(s_3) < get\_price(s_4)) \land dif\_price\_range(s_2, s_3, 0.4, 0.6)$
$similar\_price\_interval(s_1, s_2, s_3, s_4) \land similar\_time\_interval(s_1, s_2, s_3, s_4)$

<u>Pattern</u> P27 (Three Falling Peaks).

Description. The pattern has three peaks, each with a price lower than the previous peaks and proportional to one another. The pattern is confirmed when the price closes below the horizontal line drawn from the lowest low price in the pattern. See Figure 3(g).

P27: $(S = \{s_1, s_2, s_3, s_4, s_5, s_6, s_7\}) \land (len(S) = 7) \land decrease(s_2, s_4, s_6) \land (get\_price(s_7) \leq min(s_3, s_5))$
$minor\_high(s_2) \land minor\_high(s_4) \land minor\_high(s_6) \land minor\_low(s_3) \land minor\_low(s_5)$
$similar\_time\_interval(s_1, s_3, s_3, s_5) \land similar\_time\_interval(s_3, s_5, s_5, s_7) \land similar\_time\_interval(s_1, s_3, s_5, s_7)$

<u>Pattern</u> P28 (Three Rising valleys).

Description. The pattern has three minor lows, each higher than the previous lows. Each valley should look similar to the last one. The pattern is confirmed when the price closes above the highest high in the pattern. See Figure 3(h).

P28: $(S = \{s_1, s_2, s_3, s_4, s_5, s_6, s_7\}) \land (len(S) = 7) \land increase(s_2, s_4, s_6) \land (get\_price(s_7) \geq max(s_3, s_5))$
$minor\_low(s_2) \land minor\_low(s_4) \land minor\_low(s_6) \land minor\_high(s_3) \land minor\_high(s_5)$
$similar\_time\_interval(s_1, s_3, s_3, s_5) \land similar\_time\_interval(s_3, s_5, s_5, s_7) \land similar\_time\_interval(s_1, s_3, s_5, s_7)$

<u>Pattern</u> P29 (Triple Bottoms).

Description. This pattern has three distinct minor lows at approximately the same price. The pattern is confirmed when the price rises above the highest high in the formation. See Figure 3(i).

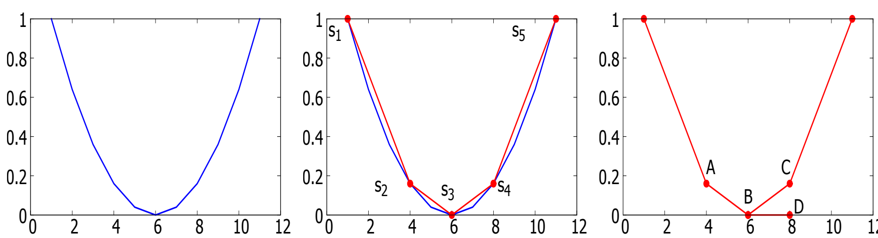
P29: $(S = \{s_1, s_2, s_3, s_4, s_5, s_6, s_7\}) \wedge (len(S) = 7) \wedge unifrom\_price(\{s_2, s_4, s_6\}) \wedge (get\_price(s_7) \geq max(s_3, s_5))$
$minor\_low(s_2) \wedge minor\_low(s_4) \wedge minor\_low(s_6) \wedge minor\_high(s_3) \wedge minor\_high(s_5)$

<u>Pattern</u> P30 (Triple Tops).

Description. "Triple Tops" has three highs, which are well separated and distinct, at approximately the same price. The pattern is confirmed when the price declines below the lowest low in the formation. See Figure 3(j).

P30: $(S = \{s_1, s_2, s_3, s_4, s_5, s_6, s_7\}) \wedge (len(S) = 7) \wedge uniform\_price(\{s_2, s_4, s_6\}) \wedge (get\_price(s_7) \leq min(s_3, s_5))$
$minor\_high(s_2) \wedge minor\_high(s_4) \wedge minor\_high(s_6) \wedge minor\_low(s_3) \wedge minor\_low(s_5)$

4.3. C3-Curved patterns

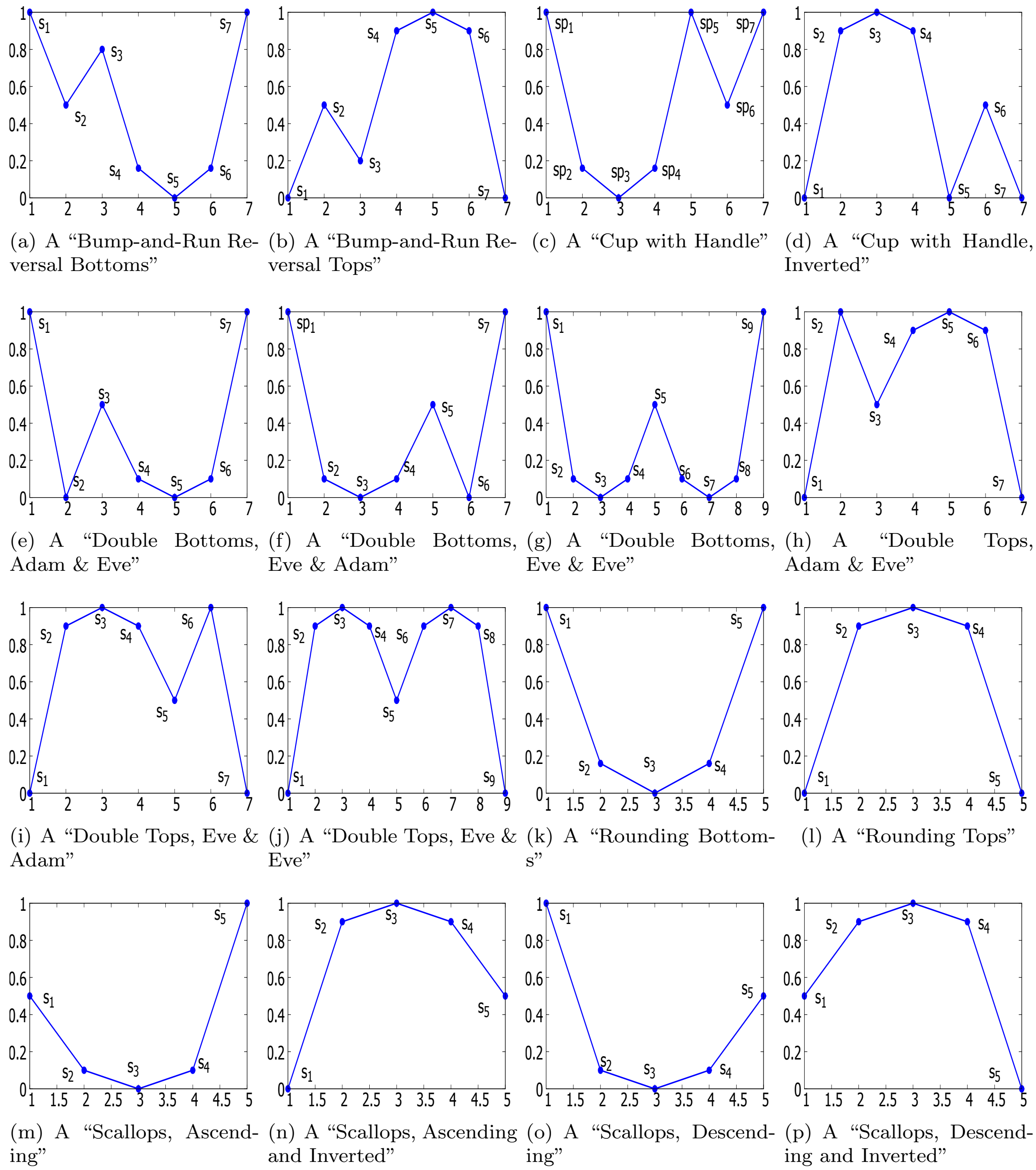


(a) (b) (c)

Figure 4: (a) A U shape. (b) The curved part is represented by three points (s_2, s_3 and s_4). (c) $\angle ABD$ is the angle of inclination of the line connecting s_2 and s_3. $\angle CBD$ is the angle of inclination of the line connecting s_3 and s_4.

In these patterns, the price decreases and increases and then increases and decreases slowly to form a curve (a U shape or inverted U shape). Figure 4(a) shows a U shape. We use three salient points to describe the curved parts of the chart patterns in the following definitions. The definition of the formation of a curve is that if the slopes of the lines connecting the salient points fall into a defined range, then a curve is formed.

For example, as shown in Figure 4(b), a curve forms when the slope of the line connecting the second salient point (s_2) and the third salient point (s_3) is less than 0 ($\tan 0°$) and greater than -1 ($\tan 135°$), and the slope of the line connecting the third salient point (s_3) and the fourth salient point (s_4) is less than 1 ($\tan 45°$) and greater than 0 ($\tan 0°$). These definitions restrict the two lines that form the curve from getting much steeper. As shown in Figure 4(c), the angle of inclination of the line connecting the second and the third salient points ($\angle ABD$) should be greater than 135° and the angle of inclination of the line connecting the third and the fourth salient points ($\angle CBD$) should be less than 45°. These definitions denote that the price decreases slowly and then increases slowly to form a curve.



(a) A "Bump-and-Run Reversal Bottoms" (b) A "Bump-and-Run Reversal Tops" (c) A "Cup with Handle" (d) A "Cup with Handle, Inverted"

(e) A "Double Bottoms, Adam & Eve" (f) A "Double Bottoms, Eve & Adam" (g) A "Double Bottoms, Eve & Eve" (h) A "Double Tops, Adam & Eve"

(i) A "Double Tops, Eve & Adam" (j) A "Double Tops, Eve & Eve" (k) A "Rounding Bottoms" (l) A "Rounding Tops"

(m) A "Scallops, Ascending" (n) A "Scallops, Ascending and Inverted" (o) A "Scallops, Descending" (p) A "Scallops, Descending and Inverted"

Figure 5: Example patterns in C3.

<u>Pattern</u> *P31 (Bump-and-Run Reversal Bottoms).*

Description. The pattern resembles a frying pan with a handle on the left sloping downward to the pan base. Then, the price levels out and finally soars out of the right side. The duration of the handle should be at least 1 month. The handle forms a down-sloping trend line called a "handle-form" trend line (the line connecting s_1 and s_3 in Figure 5(a)) that is approximately less than 45°. The down-sloping hand-form trend line (the line connecting s_1 and s_5 in Figure 5(a)) deepens to 45° to 60° or more when forming a bump. The price moves higher after the bump phase. The price, which pierces the handle-form trend line, leads to an upward breakout. The bump height (the vertical distance between the original handle-form trend line and the lowest low in the bump) should be at least twice the lead-in height (the widest vertical distance from the handle-form trend line to the daily low in the handle).

$$\textbf{P31:}\ (S = \{s_1, s_2, s_3, s_4, s_5, s_6, s_7\}) \wedge (len(S) = 7) \wedge minor\_low(s_5) \wedge (dif\_time(s_1, s_3) \geq 30)$$
$$(get\_price(s_1) > max(s_2, s_3, s_4, s_5, s_6)) \wedge (get\_price(s_3) > max(s_2, s_4, s_5, s_6)) \wedge (get\_price(s_7) > max(s_4, s_5, s_6))$$
$$(-1 \leq slope(s_1, s_3) \leq 0) \wedge (slope(s_1, s_5) \leq -1)$$
$$(-1 \leq slope(s_4, s_5) \leq 0) \wedge (0 \leq slope(s_5, s_6) \leq 1)$$
$$2 \times vd(s_1, s_2, s_3) \leq vd(s_1, s_5, s_3)$$

<u>Pattern</u> *P32 (Bump-and-Run Reversal Tops).*

Description. The pattern resembles an inverted frying pan with a handle on the left sloping upward to the pan base. Then, the price levels out and finally falls through the right side. The duration of the handle should be at least 1 month. The handle forms an up-sloping, handle-form trend line (the line connecting s_1 and s_3 in Figure 5(b)) that is usually between 30° and 45°. The up-sloping handle-form trend line (the line connecting s_1 and s_5 in Figure 5(b)) increases to 45° to 60° or more when forming a bump. The price moves lower after the bump phase. The price, which pierces the handle-form trend line, forms a downward breakout. The bump-in height (the vertical distance between the original handle-form trend line and the highest high in the bump) should be at least twice the lead-in height (the widest vertical distance from the handle-form trend line to the highest high in the handle).

$$\textbf{P32:}\ (S = \{s_1, s_2, s_3, s_4, s_5, s_6, s_7\}) \wedge (len(S) = 7) \wedge minor\_high(s_5) \wedge (dif\_time(s_1, s_3) \geq 30)$$
$$(get\_price(s_1) < min(s_2, s_3, s_4, s_5, s_6)) \wedge (get\_price(s_3) < min(s_2, s_4, s_5, s_6)) \wedge (get\_price(s_7) < min(s_4, s_5, s_6))$$
$$(0 \leq slope(s_1, s_3) \leq 1) \wedge (slope(s_1, s_5) \geq 1)$$
$$(0 \leq slope(s_4, s_5) \leq 1) \wedge (-1 \leq slope(s_5, s_6) \leq 0)$$
$$2 \times vd(s_1, s_2, s_3) \leq vd(s_1, s_5, s_3)$$

<u>Pattern</u> *P33 (Cup with Handle).*

Description. "Cup with Handle" is a U-shaped cup with a handle on the right side. The cup lips are at approximately the same price. The duration of the cup should be 7 to 65 weeks. The left lip and right lip of the cup form a horizontal trend line. The handle should form in the upper half of cup. The duration of the handle (from the right cup lip to the breakout) should be at least 1 week. The price, which pierces the horizontal trend line, leads to an upward breakout. See Figure 5(c).

P33: $(S = \{s_1, s_2, s_3, s_4, s_5, s_6, s_7\}) \wedge (len(S) = 7) \wedge similar\_price(s_1, s_5) \wedge minor\_low(s_3) \wedge minor\_low(s_6)$
$(get\_price(s_1){>}max(s_2, s_3, s_4)) \wedge (get\_price(s_5){>}max(s_2, s_3, s_4))$
$vd(s_1, s_6, s_5) \leq 0.5 \times vd(s_1, s_3, s_5)$
$(0 \leq slope(s_3, s_4) \leq 1) \wedge (-1 \leq slope(s_2, s_3) \leq 0)$
$(49 \leq dif\_time(s_1, s_5) \leq 455) \wedge (7 \leq dif\_time(s_5, s_7))$

<u>Pattern</u> *P34 (Cup with Handle, Inverted).*

Description. This pattern resembles an inverted U-shaped cup with a handle on the right side. The cup lips are at approximately the same price. The left lip and right lip of the cup form a horizontal trend line. The price of the handle must not climb above the cup (three most frequent retraces 42%, 35% and 60%). The median of the duration of the handle (from the right cup lip to the breakout) is 40 days. The price, which pierces the horizontal trend line, forms a downward breakout. See Figure 5(d).

P34: $(S = \{s_1, s_2, s_3, s_4, s_5, s_6, s_7\}) \wedge (len(S) = 7) \wedge similar\_price(s_1, s_5) \wedge minor\_high(s_3) \wedge minor\_high(s_6)$
$(get\_price(s_1){<}min(s_2, s_3, s_4)) \wedge (get\_price(s_5){<}min(s_2, s_3, s_4))$
$vd(s_1, s_6, s_5) \leq 0.5 \times vd(s_1, s_3, s_5)$
$(0 \leq slope(s_2, s_3) \leq 1) \wedge (-1 \leq slope(s_3, s_4) \leq 0)$

<u>Pattern</u> *P35 (Double Bottoms, Adam & Eve).*

Description. The left bottom is a narrow, V-shaped bottom. The right bottom appears rounded and wider. The two bottoms have similar prices. A rise between the bottoms from the lowest valley should be at least 10%. The separation between the two bottoms is at least a few (2 to 6) weeks. The confirmation line is a horizontal line drawn by the highest high between two bottoms. The price must close above the confirmation line. See Figure 5(e).

P35: $(S = \{s_1, s_2, s_3, s_4, s_5, s_6, s_7\}) \wedge (len(S) = 7) \wedge (14 \leq dif\_time(s_2, s_5) \leq 42)$
$(get\_price(s_3){>}max(s_4, s_5, s_6)) \wedge (get\_price(s_7){>}max(s_4, s_5, s_6))$
$minor\_low(s_2) \wedge minor\_low(s_5) \wedge (get\_price(s_7) \geq get\_price(s_3))$
$similar\_price(s_2, s_5) \wedge \neg similar\_price(s_3, min(s_2, s_5))$
$(0 \leq slope(s_5, s_6) \leq 1) \wedge (-1 \leq slope(s_4, s_5) \leq 0)$

<u>Pattern</u> *P36 (Double Bottoms, Eve & Adam).*

Description. The right bottom is a narrow, V-shaped bottom. The left bottom appears rounded and wider. The two bottoms have similar prices. A rise between the bottoms from the lowest valley should be at least 10%. The separation between the two bottoms is a few (2 to 7) weeks. The confirmation line is a horizontal line drawn by the highest high between the two bottoms. The price must close above the confirmation line. See Figure 5(f).

$\mathbf{P36:}$ $(S = \{s_1, s_2, s_3, s_4, s_5, s_6, s_7\}) \wedge (len(S) = 7) \wedge (14 \leq dif\_time(s_3, s_6) \leq 49)$
$(get\_price(s_1) > max(s_2, s_5, s_3, s_4)) \wedge (get\_price(s_5) > max(s_2, s_5, s_3, s_4))$
$minor\_high(s_3) \wedge minor\_high(s_6) \wedge (get\_price(s_7) \geq get\_price(s_5))$
$similar\_price(s_3, s_6) \wedge \neg similar\_price(s_5, min(s_3, s_6))$
$(0 \leq slope(s_3, s_4) \leq 1) \wedge (-1 \leq slope(s_2, s_3) \leq 0)$

<u>Pattern</u> *P37 (Double Bottoms, Eve & Eve).*

Description. The two bottoms are rounded and wide. The bottom to bottom price variation is small. A rise between the bottoms from the lowest valley should be at least 10%. The duration between the two bottoms is at least a few (2 to 7) weeks apart. The confirmation line is a horizontal line drawn by the highest high between the two bottoms. The price must close above the confirmation line. See Figure 5(g).

$\mathbf{P37:}$ $(S = \{s_1, s_2, s_3, s_4, s_5, s_6, s_7, s_8, s_9\}) \wedge (len(S) = 9) \wedge (14 \leq dif\_time(s_3, s_7) \leq 49)$
$(get\_price(s_1) > max(s_2, s_3, s_4)) \wedge (get\_price(s_5) > max(s_2, s_3, s_4))$
$(get\_price(s_5) > max(s_6, s_7, s_8)) \wedge (get\_price(s_9) > max(s_6, s_7, s_8))$
$minor\_low(s_3) \wedge minor\_low(s_7) \wedge (get\_price(s_9) \geq get\_price(s_5))$
$similar\_price(s_3, s_7) \wedge \neg similar\_price(s_5, min(s_3, s_7))$
$(0 \leq slope(s_3, s_4) \leq 1) \wedge (-1 \leq slope(s_2, s_3) \leq 0)$
$(0 \leq slope(s_7, s_8) \leq 1) \wedge (-1 \leq slope(s_6, s_7) \leq 0)$

<u>Pattern</u> *P38 (Double Tops, Adam & Eve).*

Description. The first peak is an Adam peak, which resembles an inverted V shape. The Eve peak is wider and more rounded. The top-to-top price variation is small. A dip between the two tops is usually 10% to 20% (allows variations). At least a few weeks separate the two tops (usually 2 to 7 weeks). The confirmation line is a horizontal line drawn by the lowest low between the two tops. The price must close below the confirmation line. See Figure 5(h).

$\mathbf{P38:}$ $(S = \{s_1, s_2, s_3, s_4, s_5, s_6, s_7\}) \wedge (len(S) = 7) \wedge (14 \leq dif\_time(s_2, s_5) \leq 49)$
$minor\_high(s_2) \wedge minor\_low(s_5) \wedge (get\_price(s_7) \leq get\_price(s_3))$
$(get\_price(s_3) < min(s_4, s_5, s_6)) \wedge (get\_price(s_7) < min(s_4, s_5, s_6))$
$similar\_price(s_2, s_5) \wedge \neg similar\_price(s_3, max(s_2, s_5))$
$(0 \leq slope(s_4, s_5) \leq 1) \wedge (-1 \leq slope(s_5, s_6) \leq 0)$

<u>Pattern</u> *P39 (Double Tops, Eve & Adam).*

Description. The second peak is an Adam peak, which resembles an inverted V shape. The Eve peak (first top) is wider and more rounded. The top-to-top price variation is small. A dip between the two tops is usually 10% to 25% (allows variations). The separation between the two tops is at least a few (usually 2 to 6) weeks apart. The confirmation line is a horizontal line drawn by the lowest low between the two tops. The price must close below the confirmation line. See Figure 5(i).

$$
\begin{aligned}
&\textbf{P39:}\ (S = \{s_1, s_2, s_3, s_4, s_5, s_6, s_7\}) \wedge (len(S) = 7) \wedge (14 \leq dif\_time(s_3, s_6) \leq 42) \\
&minor\_high(s_3) \wedge minor\_low(s_6) \wedge (get\_price(s_7) \leq get\_price(s_5)) \\
&(get\_price(s_1) < min(s_2, s_3, s_4)) \wedge (get\_price(s_5) < min(s_2, s_3, s_4)) \\
&similar\_price(s_3, s_6) \wedge \neg similar\_price(s_5, max(s_3, s_6)) \\
&(0 \leq slope(s_2, s_3) \leq 1) \wedge (-1 \leq slope(s_3, s_4) \leq 0)
\end{aligned}
$$

Pattern P40 (Double Tops, Eve & Eve).

Description. Both Eve peaks of "Double Tops", Eve and Eve appear rounded and wide and are not made of a single, narrow price spike. The top-to-top price variation is small. A dip between the two tops is usually 10% to 20% (allows variations). The duration between the two tops is at least a few (usually 2 to 6) weeks apart. The confirmation line is a horizontal line drawn by the lowest low between the two tops. The price must close below the confirmation line. See Figure 5(j).

$$
\begin{aligned}
&\textbf{P40:}\ (S = \{s_1, s_2, s_3, s_4, s_5, s_6, s_7, s_8, s_9\}) \wedge (len(S) = 9) \wedge (14 \leq dif\_time(s_3, s_7) \leq 42) \\
&(get\_price(s_1) < min(s_2, s_3, s_4)) \wedge (get\_price(s_5) < min(s_2, s_3, s_4)) \\
&(get\_price(s_5) < min(s_6, s_7, s_8)) \wedge (get\_price(s_9) < min(s_6, s_7, s_8)) \\
&minor\_high(s_3) \wedge minor\_high(s_7) \wedge (get\_price(s_9) \leq get\_price(s_5)) \\
&similar\_price(s_3, s_7) \wedge \neg similar\_price(s_5, max(s_3, s_7)) \\
&(0 \leq slope(s_2, s_3) \leq 1) \wedge (-1 \leq slope(s_3, s_4) \leq 0) \\
&(0 \leq slope(s_6, s_7) \leq 1) \wedge (-1 \leq slope(s_7, s_8) \leq 0)
\end{aligned}
$$

Pattern P41 (Rounding Bottoms).

Description. The price trend curves gently to construct a saucer or bowl shape. It is better to use a weekly chart to find this pattern. The right rim and the left rim of the bowl should be at approximately the same price level. The price close above the pattern end (right rim) signals an upward breakout. See Figure 5(k).

$$
\begin{aligned}
&\textbf{P41:}\ (S = \{s_1, s_2, s_3, s_4, s_5\}) \wedge (len(S) = 5) \wedge minor\_low(s_3) \wedge similar\_price(s_1, s_5) \\
&(get\_price(s_1) > max(s_2, s_3, s_4)) \wedge (get\_price(s_5) > max(s_2, s_3, s_4)) \\
&(0 \leq slope(s_3, s_4) \leq 1) \wedge (-1 \leq slope(s_2, s_3) \leq 0)
\end{aligned}
$$

Pattern P42 (Rounding Tops).

Description. "Rounding Tops" is a gentle rounding turn, similar to a half-moon or inverted U. The right rim and the left rim of the bowl should be at approximately the same price level. A close above the highest high in the pattern signals an upward breakout. A close below the pattern end (right rim) signals a downward breakout. See Figure 5(l).

P42: $(S = \{s_1, s_2, s_3, s_4, s_5\}) \wedge (len(S) = 5) \wedge minor\_high(s_3) \wedge similar\_price(s_1, s_5)$
$(get\_price(s_1) < min(s_2, s_3, s_4)) \wedge (get\_price(s_5) < min(s_2, s_3, s_4))$
$(0 \leq slope(s_2, s_3) \leq 1) \wedge (-1 \leq slope(s_3, s_4) \leq 0)$

<u>Pattern</u> *P43 (Scallops, Ascending).*

Description. The price peaks, retraces, curves around and then climbs to form a higher peak. The shape resembles the letter J. The start is a rising price trend (rarely a declining trend). It signals an upward breakout when the price closes above the highest high in the pattern or it signals a downward breakout when the price closes below the lowest low in the pattern. See Figure 5(m).

P43: $(S = \{s_1, s_2, s_3, s_4, s_5\}) \wedge (len(S) = 5) \wedge minor\_low(s_3) \wedge (get\_price(s_5) > get\_price(s_1))$
$(get\_price(s_1) > max(s_2, s_3, s_4)) \wedge (get\_price(s_5) > max(s_2, s_3, s_4))$
$dif\_price(s_5, s_3) \geq 2 \times dif\_price(s_1, s_3)$
$(0 \leq slope(s_3, s_4) \leq 1) \wedge (-1 \leq slope(s_2, s_3) \leq 0)$

<u>Pattern</u> *P44 (Scallops, Ascending and Inverted).*

Description. The shape resembles the letter J flipped upside down. The rounded top portion of the pattern usually retraces approximately 50%. The price must close above the highest high in the pattern without first falling lower than the pattern's lowest low. The breakout is upward. See Figure 5(n).

P44: $(S = \{s_1, s_2, s_3, s_4, s_5\}) \wedge (len(S) = 5) \wedge minor\_high(s_3) \wedge (get\_price(s_5) > get\_price(s_1))$
$(get\_price(s_1) < min(s_2, s_3, s_4)) \wedge (get\_price(s_5) < min(s_2, s_3, s_4))$
$dif\_price(s_1, s_3) \geq 2 \times dif\_price(s_5, s_3)$
$(0 \leq slope(s_2, s_3) \leq 1) \wedge (-1 \leq slope(s_3, s_4) \leq 0)$

<u>Pattern</u> *P45 (Scallops, Descending).*

Description. The pattern resembles the letter J reversed. The end (right peak) usually retraces 60%. The breakout is upward or downward depending on whether the price closes above or below the highest high or lowest low in the pattern. See Figure 5(o).

P45: $(S = \{s_1, s_2, s_3, s_4, s_5\}) \wedge (len(S) = 5) \wedge minor\_low(s_3) \wedge (get\_price(s_5) < get\_price(s_1))$
$(get\_price(s_1) > max(s_2, s_3, s_4)) \wedge (get\_price(s_5) > max(s_2, s_3, s_4))$
$dif\_price(s_1, s_3) \geq 2 \times dif\_price(s_5, s_3)$
$(0 \leq slope(s_3, s_4) \leq 1) \wedge (-1 \leq slope(s_2, s_3) \leq 0)$

<u>Pattern</u> *P46 (Scallops, Descending and Inverted).*

Description. The shape resembles an upside-down J. The start trend is downward. From the scallop start to its high is usually approximately half of the length of the following downward move. The price must close below the lowest low in the pattern. See Figure 5(p).

P46: $(S = \{s_1, s_2, s_3, s_4, s_5\}) \wedge (len(S) = 5) \wedge minor\_high(s_3) \wedge (get\_price(s_5) < get\_price(s_1))$
$(get\_price(s_1) < min(s_2, s_3, s_4)) \wedge (get\_price(s_5) < min(s_2, s_3, s_4))$
$dif\_price(s_5, s_3) \geq 2 \times dif\_price(s_1, s_3)$
$(0 \leq slope(s_2, s_3) \leq 1) \wedge (-1 \leq slope(s_3, s_4) \leq 0)$

4.4. C4-Candlestick patterns with spikes

Definition 6 (Sequence of candlesticks to be classified) $C = \{c_1, c_2, ..., c_n\}$ is a sequence of candlesticks to be classified as a chart pattern or not. $c_i = (s_H, s_L)$ is a candlestick that is represented by the point of the highest price and the point of the lowest price. $len(C)$ returns the length (the number of candlesticks) of a chart pattern. $C[c_i, c_j]$ denotes the subsequence of C from candlestick c_i to c_j.

Function 28 Functions to access the elements of C. $get\_high(c_i)$ returns the highest point of candlestick c_i. $get\_low(c_i)$ returns the lowest point of candlestick c_i. $get\_time\_cand(c_i)$ returns the time of candlestick c_i. $get\_high\_price(c_i)$ returns the price of the highest point of candlestick c_i. $get\_low\_price(c_i)$ returns the price of the lowest point of candlestick c_i.
$get\_high(c_i) = s_H, get\_low(c_i) = s_L,$
$get\_time\_cand(c_i) = get\_time(s_H), get\_high\_price(c_i) = get\_price(s_H), get\_low\_price(c_i) = get\_price(s_L).$

Definition 7 (Sequence of candlesticks preceding C) $B = \{b_1, b_2, ..., b_n\}$ is a sequence of candlesticks preceding sequence C.

Definition 8 (Subsequence of B) $subq(B, n)$ is a subsequence of B and its length is at most n. B is a sequence of candlesticks preceding sequence C.

$$subq(B, n) = \begin{cases} B[b_{get\_time\_cand(c_1)-n}, b_{get\_time\_cand(c_1)-1}], & len(B) \geq n \\ B, & len(B) < n \end{cases}$$

Function 29 $spike\_down(c_i)$ returns true if the candlestick, c_i, is a downward spike.
$spike\_down(c_i) \equiv get\_low\_price(c_i) > get\_low\_price(c_{i+1}) \wedge get\_low\_price(c_i) > get\_low\_price(c_{i-1}).$

Function 30 $spike\_up(c_i)$ returns true if the candlestick, c_i, is an upward spike.
$spike\_up(c_i) \equiv get\_high\_price(c_i) > get\_high\_price(c_{i+1}) \wedge get\_high\_price(c_i) > get\_high\_price(c_{i-1}).$

Function 31 $spike\_len\_down(c_i)$ returns the length of downward spike c_i.
$spike\_len\_down(c_i) = min(dif\_price(get\_low(c_i), get\_low(c_{i-1})), dif\_price(get\_low(c_i), get\_low(c_{i+1}))), spike\_down(c_i)$

Function 32 $spike\_len\_up(c_i)$ returns the length of upward spike c_i.
$spike\_len\_up(c_i) = min(dif\_price(get\_high(c_i), get\_high(c_{i-1})), dif\_price(get\_high(c_i), get\_high(c_{i+1}))), spike\_up(c_i)$

Function 33 $average\_len\_down(C)$ returns the average length of the downward spikes extracted from spike sequence C.

$average\_len\_down(C) = \frac{\sum(spike\_len\_down(c_i))}{len(C)}, c_i \in C \wedge spike\_down(c_i).$

Function 34 $average\_len\_up(C)$ returns the average length of the upward spikes extracted from the spike sequence C.

$average\_len\_up(C) = \frac{\sum(spike\_len\_up(c_i))}{len(C)}, c_i \in C \wedge spike\_up(c_i).$

Definition 9 (Gaps). $gap\_down(c_i, c_{i+1})$ is a gap that forms when the lowest price of c_i is greater than the highest price of c_{i+1}. $gap\_up(c_i, c_{i+1})$ is a gap that forms when the highest price of c_i is less than the lowest price of c_{i+1}. We use $gap(c_i, c_{i+1})$ to unify these two kinds of gaps.

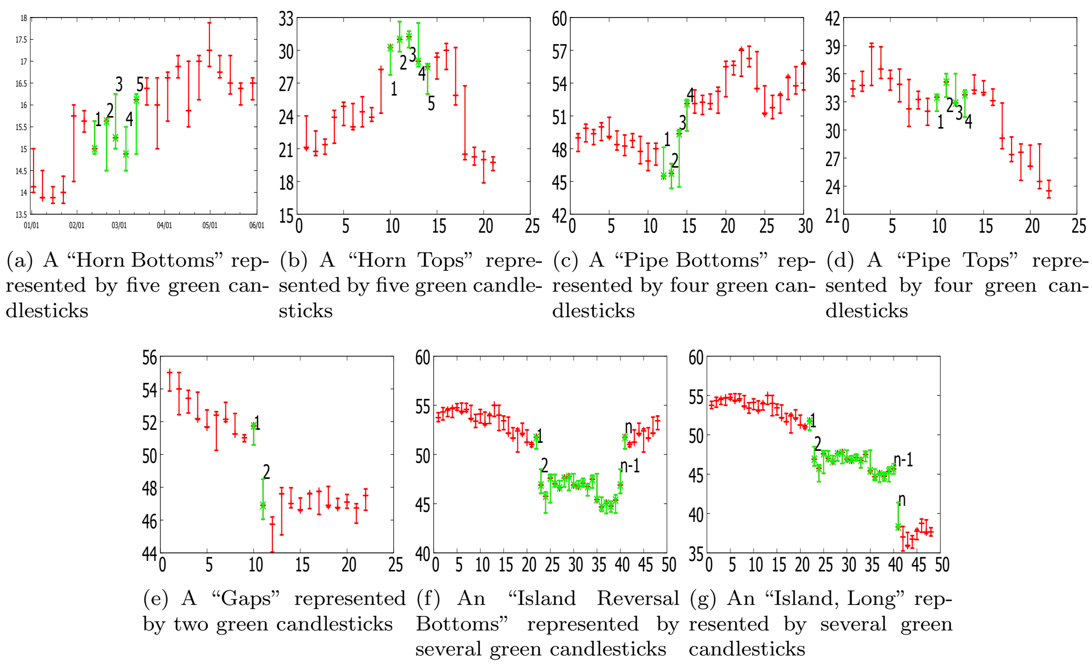
$gap\_down(c_i, c_{i+1}) \equiv get\_low\_price(c_i) > get\_high\_price(c_{i+1}),$

$gap\_up(c_i, c_{i+1}) \equiv get\_high\_price(c_i) < get\_low\_price(c_{i+1}),$

$gap(c_i, c_{i+1}) \equiv (get\_low\_price(c_i) > get\_high\_price(c_{i+1})) \mid (get\_high\_price(c_{i+1}) < get\_low\_price(c_{i+1})).$

Function 35 $gap\_width(c_i, c_{i+1})$ returns the width of a gap.

$$gap\_width(c_i, c_{i+1}) = \begin{cases} dif\_price(get\_low\_price(c_i), get\_high\_price(c_{i+1})), & get\_low\_price(c_i) > get\_high\_price(c_{i+1}) \\ dif\_price(get\_high\_price(c_i), get\_low\_price(c_{i+1})), & get\_high\_price(c_i) < get\_low\_price(c_{i+1}) \end{cases}$$



(a) A "Horn Bottoms" represented by five green candlesticks

(b) A "Horn Tops" represented by five green candlesticks

(c) A "Pipe Bottoms" represented by four green candlesticks

(d) A "Pipe Tops" represented by four green candlesticks

(e) A "Gaps" represented by two green candlesticks

(f) An "Island Reversal Bottoms" represented by several green candlesticks

(g) An "Island, Long" represented by several green candlesticks

Figure 6: Example patterns in C4 and C5.

<u>Pattern</u> *P47 (Horn Bottoms).*

Description. This pattern is formed by five weekly candlesticks. "Horn Bottoms" has two downward spikes separated by a 1-week duration. In this pattern, at least two spikes should be longer than similar spikes over the previous year (approximately 48 weeks in weekly data). The pattern is confirmed when the price (i.e., the highest price of Stick 5) closes above the highest high in the previous three sticks. See Figure 6(a).

P47: $(len(C) = 5) \wedge similar\_price(get\_low(c_2), get\_low(c_4)) \wedge spike\_dwon(c_2) \wedge spike\_down(c_4)$
$spike\_len\_down(c_2){>}average\_len\_down(subq(B, 48))$
$spike\_len\_down(c_4){>}average\_len\_down(subq(B, 48))$
$get\_high\_price(c_5){>}max(get\_high\_price(c_2), get\_high\_price(c_3), get\_high\_price(c_4))$

<u>Pattern</u> *P48 (Horn Tops).*

Description. "Horn Tops" is formed by five weekly candlesticks. The pattern has two upward price spikes separated by 1 week on the weekly chart. The two spikes should be longer than similar spikes over the previous year (approximately 48 weeks in weekly data). The pattern is confirmed when the price (i.e., the lowest price of Stick 5) closes below the lowest low in the pattern. See Figure 6(b).

P48: $(len(C) = 5) \wedge similar\_price(get\_high(c_2), get\_high(c_4)) \wedge spike\_up(c_2) \wedge spike\_up(c_4)$
$spike\_len\_up(c_2){>}average\_len\_up(subq(B, 48))$
$spike\_len\_up(c_4){>}average\_len\_up(subq(B, 48))$
$get\_low\_price(c_5){<}min(get\_low\_price(c_2), get\_low\_price(c_3), get\_low\_price(c_4))$

<u>Pattern</u> *P49 (Pipe Bottoms).*

Description. This pattern is formed by four weekly candlesticks. The pattern has two adjacent downward price spikes on the weekly chart. The two pipes have a large price overlap. The length of the two pipes should be well above the average length of the downward spikes over the previous year (approximately 48 weeks in weekly data). The pattern is confirmed as being "Pipe Bottoms" when the price closes above the highest high in the pattern before dropping below the lowest low. See Figure 6(c).

P49: $len(C) = 4 \wedge similar\_price(get\_low(c_2), get\_low(c_3))$
$get\_low\_price(c_2){<}min(get\_low\_price(c_1), get\_low\_price(c_4))$
$get\_low\_price(c_3){<}min(get\_low\_price(c_1), get\_low\_price(c_4))$
$dif\_price(get\_low(c_2), get\_low(c_1)){>}average\_len\_down(subq(B, 48))$
$dif\_price(get\_low(c_3), get\_low(c_4)){>}average\_len\_down(subq(B, 48))$
$get\_high\_price(c_4){>}max(get\_high\_price(c_2), get\_high\_price(c_3))$

<u>Pattern</u> *P50 (Pipe Tops).*

Description. This pattern is formed by four weekly candlesticks. The pattern has two adjacent upward price spikes on the weekly chart. The two spikes should be longer than similar spikes over the previous year (approximately 48 weeks in weekly data). The two spikes should have a large price overlap. The price must close below the lowest low in the pattern before rising above the top. See Figure 6(d).

P50: $len(C) = 4 \wedge similar\_price(get\_high(c_2), get\_high(c_3))$

$get\_high\_price(c_2) > max(get\_high\_price(c_1), get\_high\_price(c_4))$

$get\_high\_price(c_3) > max(get\_high\_price(c_1), get\_high\_price(c_4))$

$dif\_price(get\_high(c_2), get\_high(c_1)) > average\_len\_down(subq(B, 48))$

$dif\_price(get\_high(c_3), get\_high(c_4)) > average\_len\_down(subq(B, 48))$

$get\_low\_price(c_4) < min(get\_low\_price(c_2), get\_low\_price(c_3))$

4.5. C5-Candlestick patterns with gaps

Pattern P51 (Gaps).

Description. The "Gaps" pattern appears when the previous day's daily highest price is below the present day's lowest price or when the previous day's lowest price is above the present day's highest price as shown in Definition 9. See Figure 6(e).

P51: $len(C) = 2 \wedge gap(c_1, c_2)$

Pattern P52 (Island Reversal).

Description. There are two "Island Reversal" patterns. An "Island Reversal Tops" pattern forms when the price gaps up to the formation and then gaps down at the same price level. An "Island Reversal Bottoms" pattern forms when the price gaps down to the formation and then gaps up at the same price level. An "Island Reversal" pattern is formed by several continuous daily candlesticks separated by two gaps. Gaps set off both "Island Reversal Tops" and "Island Reversal Bottoms" and share all or part of the same price level. To identify the pattern, two gaps separating a series of continuous candlesticks should first be located. Gaps should be at least \$0.25 wide or larger. The pattern should be shorter than 4 months (approximately 120 days in daily data). See Figure 6(f).

P52-1 Island Reversal Tops:

$len(C) \geq 4 \wedge gap\_up(c_1, c_2) \wedge gap\_down(c_{n-1}, c_n)$

$\forall c_i, c_j \notin c_1, c_2, c_{n-1}, c_n \nRightarrow gap(c_i, c_j)$

$gap\_width(c_1, c_2) \geq 0.25 \wedge gap\_width(c_{n-1}, c_n) \geq 0.25$

$dif\_time(get\_high(c_1), get\_high(c_n)) < 120$

$similar\_price(get\_low(c_2), get\_low(c_{n-1}))$

P52-2 Island Reversal Bottoms:

$len(C) \geq 4 \wedge gap\_down(c_1, c_2) \wedge gap\_up(c_{n-1}, c_n)$

$\forall c_i, c_j \notin c_1, c_2, c_{n-1}, c_n \nRightarrow gap(c_i, c_j)$

$gap\_width(c_1, c_2) \geq 0.25 \wedge gap\_width(c_{n-1}, c_n) \geq 0.25$

$dif\_time(get\_high(c_1), get\_high(c_n)) < 120$

$similar\_price(get\_high(c_2), get\_high(c_{n-1}))$

Pattern P53 (Island, Long).

Description. The "Island, Long" pattern is formed by several continuous sticks that are separated by two gaps as a whole, resembling an island in a time series represented by candlesticks. An "Island, Long" pattern is formed by several continuous daily candlesticks separated by two gaps. The two gaps separating a series of continuous candlesticks should initially be located to identify the pattern. The gaps should be at least \$1 wide and the pattern should be shorter than 4 months (approximately 120 days in daily data). An "Island Reversal" pattern is also an "Island, Long" pattern, but an "Island, Long" is not necessarily an "Island Reversal". See Figure 6.

P53: $len(C) \geq 4 \wedge gap(c_1, c_2) \wedge gap(c_{n-1}, c_n)$
$\forall c_i, c_j \notin c_1, c_2, c_{n-1}, c_n \nRightarrow gap(c_i, c_j)$
$gap\_width(c_1, c_2) \geq 1 \wedge gap\_width(c_{n-1}, c_n) \geq 1$
$dif\_time(get\_high(c_1), get\_high(c_n)) < 120$

5. Experiments on real datasets

The real datasets used in the experiment were downloaded from Yahoo Finance [25]. For patterns represented by points (C1, C2 and C3), we used the historical daily closing prices of the NYSE from 3 January 2005 to 31 December 2015, which contained 2,769 points. The closing price from the dataset is an important attribute, as it represents the final evaluation of the stock made by the market during the day [9]. For patterns formed by weekly candlesticks (C4), we used the historical weekly candlesticks of the NYSE from 3 January 2005 to 31 December 2015. For patterns formed by daily candlesticks (C5), we used the historical daily candlesticks of the NYSE from 3 January 2005 to 31 December 2015.

We adopted the RB approach [11] to search the chart patterns in the real datasets. All of the rules used in the RB approach were transformed from the pattern definitions given in previous sections. For the patterns formed by points (from the C1, C2 and C3 categories), we used a sliding window to obtain a subsequence from the historical price. Next, we selected a number of PIPs from the subsequence. Finally, we examined the resulting segmented subsequence with the rules translated from the definitions described above. Redundant patterns that were within a +1 or -1 difference in position were eliminated. For patterns formed by candlesticks (from the C4 and C5 categories), we traversed the whole time series and used the rules translated from the specifications to identify the patterns in the time series without using sliding window or segmentation steps.

5.1. C1-Variable fluctuation patterns

Table 1 shows the number of variable fluctuation patterns (C1) found. In this table, ♯PIPs is the number of PIPs and ♯Patterns is the number of patterns found. "NIL" indicates that we did not find the pattern in the stock time series. The window sizes (WSs) were set to 30 for most of the patterns of C1, except for "Flags" (P9), "Flags, High and Tight" (P10) and "Pennants" (P13), which have a short duration that is less than 3 weeks. Their WSs were set to 21.

To find all of the variations of the patterns in C1, we traversed all of the possible numbers of PIPs when the WS was fixed for each of the chart patterns in C1. For example, "Broadening Bottoms" (P1) has at least four touches so the least number of PIPs is six. The largest number of PIPs was set to the WS (i.e., 30 in this case). We gradually increased from 6 PIPs to 30 PIPs to find "Broadening Bottoms".

In the experiment, "Head-and-Shoulders Bottoms, Complex" and "Head-and-Shoulders Tops, Complex" patterns were not found in the NYSE dataset (see Table 1). To verify the correctness of

Table 1: The number of variable fluctuation patterns (C1) found in the NYSE.

Name	[♯PIPs,♯Patterns]
P1. Broadening Bottoms	[6,20],[7,6]
P2. Broadening Formations, Right-Angled and Ascending	[6,14],[7,7][9,1]
P3. Broadening Formations, Right-Angled and Descending	[6,19],[7,8],[8,3]
P4. Broadening Tops	[6,22],[7,8],[8,1]
P5. Broadening Wedges, Ascending	[8,13],[9,10],[10,6],[11,4],[12,3],[13,2],[14,1]
P6. Broadening Wedges, Descending	[6,30],[7,18],[8,7],[9,5],[10,3],[11,1],[12,2],[13,1]
P7. Diamond Bottoms	[8,9],[9,1],[10,3],[11,1],[12,1]
P8. Diamond Tops	[8,8],[9,2],[10,2],[14,1]
P9. Flags	[6,15],[7,4]
P10. Flags, High and Tight	[6,7],[7,4]
P11. Head-and-Shoulders Bottoms, Complex	NIL
P12. Head-and-Shoulders Tops, Complex	NIL
P13. Pennants	[6,36],[7,60],[8,31],[9,15],[10,11],[11,4],[12,3],[13,2] [14,1],[15,1]
P14. Rectangle Bottoms	[6,7],[7,3],[8,1]
P15. Rectangle Tops	[6,4],[7,2]
P16. Triangle, Ascending	[6,22],[7,8]
P17. Triangle, Descending	[6,12],[7,5],[8,2],[9,1]
P18. Triangle, Symmetrical	[6,33],[7,8],[8,2],[9,1],[10,1]
P19. Wedges, Falling	[7,17],[8,8],[9,4],[10,1],[11,2],[12,1],[13,1]
P20. Wedge, Rising	[7,25],[8,19],[9,12],[10,7],[11,6],[12,1],[13,2]

our definitions for these two patterns, we performed an additional experiment using the historical daily closing prices of the HANG SENG INDEX (HSI) from 3 January 2005 to 31 December 2015, which contained 2,759 points. In the HSI dataset, we found one "Head-and-Shoulders Bottoms, Complex" and two "Head-and-Shoulders Tops, Complex" patterns. Figure 7(a) shows a "Head-and-Shoulders Bottoms, Complex" with one head and four shoulders found in the HSI. Figure 7(b) shows a "Head-and-Shoulders Tops, Complex" with three heads and two shoulders found in the HSI.

Table 2: The number of fixed fluctuation patterns (C2) found in the NYSE.

Name	[♯PIPs,♯Patterns]	Name	[♯PIPs,♯Patterns]
21. Double Bottoms, Adam & Adam	[5,3]	26. Measured Move Up	[4,1]
22. Double Tops, Adam & Adam	[5,2]	27. Three Falling Peaks	[7,0]
23. Head-and-Shoulders Bottoms	[7,0]	28. Three Rising Valley	[7,4]
24. Head-and-Shoulders Tops	[7,3]	29. Triple Bottoms	[7,3]
25. Measured Move Down	[4,2]	30. Triple Tops	[7,4]

5.2. C2-Fixed fluctuation patterns

Table 2 shows the number of fixed fluctuation patterns (C2) found. In this table, ♯PIPs is the number of PIPs and ♯Patterns is the number of patterns found. The WSs were set to 60 for "Double Bottoms, Adam & Adam" and "Double Tops, Adam & Adam" patterns, as the separation of two bottoms or tops is longer than 3 or 5 weeks duration. For other patterns of C2, the WSs were set to 30.

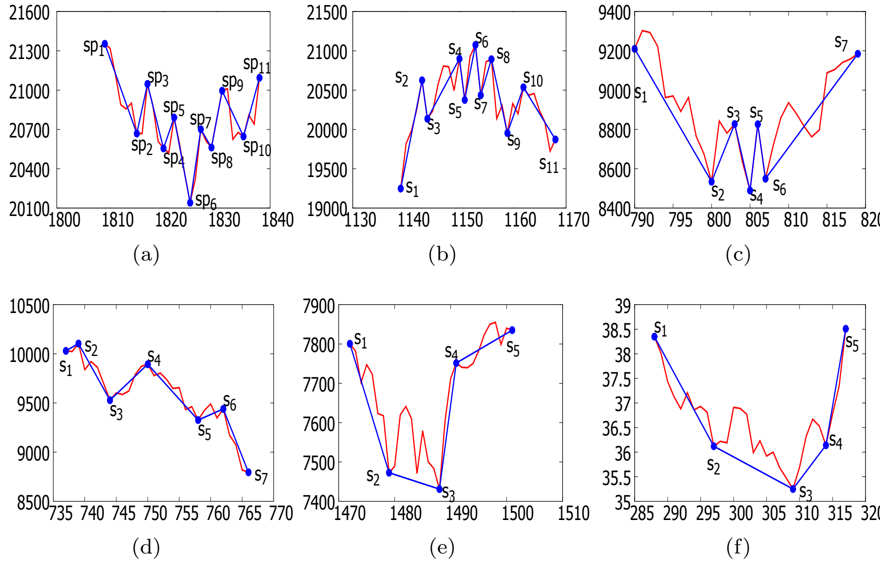


Figure 7: For all of the figures, the red lines are the original subsequences of the stocks and the blue lines are the segmentation results on the corresponding subsequence. (a) A subsequence of a "Head-and-Shoulders Bottoms, Complex" pattern was found in the HSI. (b) A subsequence of a "Head-and-Shoulders Tops, Complex" pattern was found in HSI. (c) A "Head-and-Shoulders Bottoms" pattern was found in the NYSE. (d) A "Three Falling Peaks" pattern was found in the NYSE. (e) A "Rounding Bottoms" pattern without a restriction on curve formation was found in the NYSE. (f) A "Rounding Bottoms" pattern with a restriction on curve formation was found in the Amazon.com (AMZN) dataset.

The experimental results in Table 2 reveal that no "Head-and-Shoulders Bottoms" (P23) or "Three Falling Peaks" (P27) patterns were found in the dataset. To check the correctness of the specification, an additional test was performed by removing the constraint on the similarity of the distances of both shoulders from the head from the rules of "Head-and-Shoulders Bottoms". In this additional test, we found one "Head-and-Shoulders Bottoms" pattern in the NYSE as shown in Figure 7(c). We can see that the shoulders (s_2 and s_6) and necks (s_3 and s_5) are at similar price levels, but the distance from the left shoulder (s_2) to the head (s_4) is much greater than the distance from the right shoulder (s_6) to the head (s_4). Similarly, if we remove the constraint on the proportionality of the "Three Falling Peaks" pattern, we find 13 "Three Falling Peaks" patterns. Figure 7(d) shows a "Three Falling Peaks", which has three peaks of different width or size.

5.3. C3-Curved patterns

The WSs were set to 120 for "Bump-and-Run Reversal Bottoms" and "Bump-and-Run Reversal Tops". As the minimum duration of the handles for these two patterns is 1 month, we used a bigger WS to find them. The WSs were set to 60 for "Cup with Handle" and "Cup with Handle, Inverted", as their handles should be at least 1 week and the cups should be longer than 7 weeks. The WSs for the remaining patterns of C3 were set to 30.

In this experiment, we did not find any patterns of C3. One reason is that the rules for the appearance of curves are too strict. A perfect U-shaped curve may not appear in these WSs or in the datasets used for testing. The NYSE is a stock market index covering over 2,000 stocks. The price of the NYSE index often fluctuates sharply within a short period compared to the stock price of a single company. The curves form when the price declines or increases slowly, which

Table 3: The number of candlestick patterns with spikes and candlestick patterns with gaps found in the NYSE.

Name	♯Patterns	Name	♯Patterns
47. Horn Bottoms	1	51. Gaps	184
48. Horn Tops	0	52. Island Reversal	28
49. Pipe Bottoms	11	53. Island, Long	170
50. Pipe Tops	8		

results in a gentle rounding turn akin to a half-moon or U/inverted U. We used three PIPs to represent a curve and a constraint was used to ensure the slopes of the two lines representing the curve fit into a certain range. A curve seldom appears in the time series of a stock market index, such as the NYSE. If we remove the restrictions on the slopes of the two lines forming the curve, we can find more patterns in the NYSE. For instance, seven "Rounding Bottoms" were found in the NYSE without the restrictions on the slopes of the two lines forming the curved bottom (i.e., $0 \leq slope(s_3, s_4) \leq 1, -1 \leq slope(s_2, s_3) \leq 0$ as shown in the definition of "Rounding Bottoms" (P41)). To further clarify our observation, we also used the same definition with restrictions on the slopes of the two lines forming the curved bottom to search "Rounding Bottoms" in the historical daily closing prices of AMZN from 3 January 2005 to 31 December 2015, which contains 2,769 points. We found two "Rounding Bottoms" in the AMZN dataset.

Figure 7(e) is a "Rounding Bottoms" without restrictions on the curve found in the NYSE and Figure 7(f) is a "Rounding Bottoms" with restrictions on the curve found in the AMZN dataset. As shown in Figure 7(e), the difference between the second and third PIPs is \$41.69 and the duration between these two points is 9 days. The difference between the third and fourth PIPs is \$320.64 and the duration between the two points is 3 days. In Figure 7(f) the difference between the second and third PIPs is \$0.87 and the duration between them is 12 days. The difference between the third and fourth PIPs is \$0.89 and the duration between them is 5 days. These results reveal that a single stock is more likely to satisfy the restrictions on the slopes of the two lines forming the curve than a stock market index.

5.4. C4-Candlestick patterns with spikes and C5-Candlestick patterns with gaps

Table 3 shows the number of candlestick patterns found with spikes (C4) and with gaps (C5). In this table, ♯Patterns is the number of patterns found. According to Table 3, none of the "Horn Tops" patterns were found in the NYSE dataset. Although we did not perform an extra experiment on this particular case, we believe that more "Horn Tops" may be found by relaxing some of the rules.

5.5. Experiments on different pattern matching approaches

In this experiment, we classify "Wedges, Rising"(P20-1) from C1, "Head-and-Shoulders Tops" (P24) from C2 and "Cup with Handle" patterns (P33) from C3 using the RB, template-based (TB) method [11], the pattern matching approach based on Euclidean distance (ED), and dynamic time warping (DTW). We use the historical daily closing prices of the NYSE for the experiment. A sliding window of length 30 shifts one data point at a time to get a sub-sequence from the NYSE.

Then PIP segmentation method is performed to extract perceptual important points among these 30 points. Finally, the segmented sequence is input to different pattern matching methods. In this experiment, redundant patterns that are within +1 or -1 difference in position are eliminated.

For RB, we use the rules translated from the formal specifications. For TB, ED and DTW, we design templates according to the formal specifications. TB calculates the amplitude distance and temporal distance between a template of chart pattern and the segmented sequences. ED and DTW also can be used to calculate the distance between the template and the segmented sequences. Table 4 shows the number of patterns found by the four pattern matching approaches in the NYSE.

From the results of Table 4, none of "Cup with Handle" are found by RB. As we've mentioned before that we used three PIPs to represent a curve and a constraint was used to ensure the slopes of the two lines representing the curve fit into a certain range. A curve seldom appears in the time series of a stock market index, such as the NYSE. Although TB, ED and DTW use the same templates which are designed according to the formal specifications, TB approach is more stricter than ED and DTW. TB calculates both the amplitude and temporal distance between a template and a segmented sequence, while ED and DTW calculate the amplitude distance between a template and a segmented sequence in different ways. Therefore, when compared with ED and DTW, less number of patterns are found by TB.

Table 4: The number of patterns found by four pattern matching approaches in the historical price of NYSE.

	Wedges, Rising	Head-and-Shoulders Tops	Cup with Handle
RB	20	3	0
TB	0	3	11
ED	9	25	15
DTW	16	32	23

6. Conclusions

Although a number of new approaches have been proposed in recent years to solve the chart pattern-matching problem, one of the key questions that remains unsolved is how these chart patterns should be defined in a comprehensive and unambiguous way. We classify 53 chart patterns into five categories and compile comprehensive formal specifications based on their shapes and relevant properties. Data mining algorithms can use these formal specifications for classification without significant modification. We believe that the specifications we propose can be used together as a reference model for future research in this area.

In the experiments, the formal specifications were translated into rules and the RB approach was adopted to classify the 53 chart patterns from the real dataset of the NYSE. The experimental results show that the rules translated from the specifications can be effectively used to identify chart patterns from real datasets. The experimental results also reveal a number of interesting observations. The absence of a particular pattern is often caused by the rules, which are often too strict for volatile stocks and indices. For instance, the rules for identifying a curve may be considered to be too strict for identifying patterns with curves in the NYSE, which is more likely to have sharp price movements. However, the definition of the curve performs well in stocks when there are less price fluctuations. In this regard, by relaxing or restricting conditions, traders can effectively control the expected quality and quantity of the patterns to be identified from a dataset. Therefore, our approach clearly demonstrates that the rigorous formal specifications that we have compiled can be extremely useful for traders when they are translated into rules. As for future work, we plan to develop a real-time decision support system based on the formal specifications of the chart patterns provided in this paper.

Acknowledgement

This research was funded by the Research Committee of the University of Macau, grants MYRG2015-00054-FST and MYRG2016-00148-FST.

[1] Achelis, S. B. (2001). *Technical Analysis from A to Z*. McGraw Hill New York.

[2] Anand, S., Chin, W.-N., and Khoo, S.-C. (2001). Charting patterns on price history. In *ACM SIGPLAN Notices*, volume 36, pages 134–145. ACM.

[3] Bergroth, L., Hakonen, H., and Raita, T. (2000). A survey of longest common subsequence algorithms. In *String Processing and Information Retrieval, 2000. SPIRE 2000. Proceedings. Seventh International Symposium on*, pages 39–48. IEEE.

[4] Berndt, D. J. and Clifford, J. (1994). Using dynamic time warping to find patterns in time series. In *KDD workshop*, volume 10, pages 359–370. Seattle, WA.

[5] Bulkowski, T. N. (2011). *Encyclopedia of chart patterns*. John Wiley & Sons, 2nd edition.

[6] Chen, C.-H., Tseng, V. S., Yu, H.-H., and Hong, T.-P. (2013). Time series pattern discovery by a pip-based evolutionary approach. *Soft Computing*, 17(9):1699–1710.

[7] Chung, F.-L., Fu, T.-C., Luk, R., and Ng, V. (2001). Flexible time series pattern matching based on perceptually important points. In *International joint conference on artificial intelligence workshop on learning from temporal and spatial data*, pages 1–7.

[8] Dempster, A. P., Laird, N. M., and Rubin, D. B. (1977). Maximum likelihood from incomplete data via the em algorithm. *Journal of the royal statistical society. Series B (methodological)*, 39(1):1–38.

[9] Edwards, R. D., Magee, J., and Bassetti, W. (2007). *Technical analysis of stock trends*. CRC Press.

[10] Esling, P. and Agon, C. (2012). Time-series data mining. *ACM Computing Surveys (CSUR)*, 45(1):12.

[11] Fu, T.-c., Chung, F.-l., Luk, R., and Ng, C.-m. (2007). Stock time series pattern matching: Template-based vs. rule-based approaches. *Engineering Applications of Artificial Intelligence*, 20(3):347–364.

[12] Ge, X. and Smyth, P. (2000). Deformable markov model templates for time-series pattern matching. In *Proceedings of the sixth ACM SIGKDD international conference on Knowledge discovery and data mining*, pages 81–90. ACM.

[13] Hochreiter, S. and Schmidhuber, J. (1997). Long short-term memory. *Neural computation*, 9(8):1735–1780.

[14] Keogh, E., Chu, S., Hart, D., and Pazzani, M. (2001). An online algorithm for segmenting time series. In *Data Mining, 2001. ICDM 2001, Proceedings IEEE International Conference on*, pages 289–296. IEEE.

[15] Keogh, E. J. and Pazzani, M. J. (2000). A simple dimensionality reduction technique for fast similarity search in large time series databases. In *Knowledge Discovery and Data Mining. Current Issues and New Applications*, pages 122–133. Springer.

[16] Kim, S. and Smyth, P. (2006). Segmental hidden markov models with random effects for waveform modeling. *The Journal of Machine Learning Research*, 7:945–969.

[17] LeCun, Y., Bengio, Y., and Hinton, G. (2015). Deep learning. *Nature*, 521(7553):436–444.

[18] Lin, J., Keogh, E., Wei, L., and Lonardi, S. (2007). Experiencing sax: a novel symbolic representation of time series. *Data Mining and knowledge discovery*, 15(2):107–144.

[19] Liu, S., Yang, N., Li, M., and Zhou, M. (2014). A recursive recurrent neural network for statistical machine translation.

[20] Lo, A. W., Mamaysky, H., and Wang, J. (2000). Foundations of technical analysis: Computational algorithms, statistical inference, and empirical implementation. *Journal of Finance*, 55(4):1705–1770.

[21] Mikolov, T., Karafiát, M., Burget, L., Cernockỳ, J., and Khudanpur, S. (2010). Recurrent neural network based language model. In *Interspeech*, volume 2, page 3.

[22] Si, Y.-W. and Yin, J. (2013). Obst-based segmentation approach to financial time series. *Engineering Applications of Artificial Intelligence*, 26(10):2581–2596.

[23] Struzik, Z. R. and Siebes, A. (1999). The haar wavelet transform in the time series similarity paradigm. In *Principles of Data Mining and Knowledge Discovery*, pages 12–22. Springer.

[24] Wan, Y., Gong, X., and Si, Y.-W. (2016). Effect of segmentation on financial time series pattern matching. *Applied Soft Computing*, 38:346–359.

[25] Yahoo (2009). Yahoo finance. `https://ca.finance.yahoo.com/q/hp?s=YHOO`. [Accessed: 23-Mar-2016].

[26] Zapranis, A. and Tsinaslanidis, P. (2010). Identification of the head-and-shoulders technical analysis pattern with neural networks. In *Artificial Neural Networks–ICANN 2010*, pages 130–136. Springer.

[27] Zhang, Z., Jiang, J., Liu, X., Lau, R., Wang, H., and Zhang, R. (2010). A real time hybrid pattern matching scheme for stock time series. In *Proceedings of the Twenty-First Australasian Conference on Database Technologies-Volume 104*, pages 161–170. Australian Computer Society, Inc.